
Time Series

Rennan F. B. Marujo

Édipo Cremon



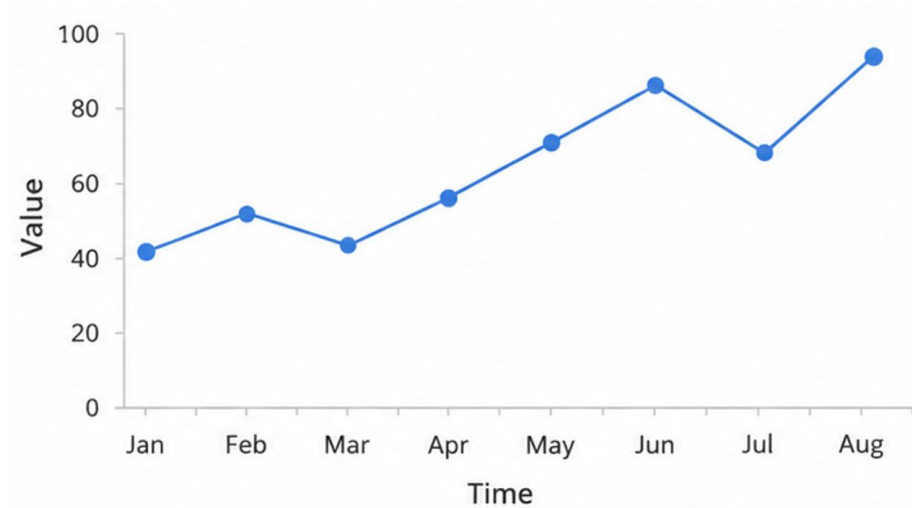
What is a Time Series?

Definition

A time series is a set of observations of a variable arranged in chronological order, allowing its evolution over time to be analyzed.

Discrete observation?

Equidistant measurements?



Regular intervals

Observations taken at equal time intervals



Irregular intervals

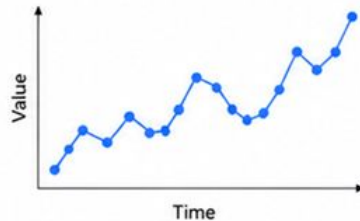
Observations taken at unequal or irregular times.

What do we want with time series?



1. Understand Behaviors

Explore how a variable evolves over time to understand its underlying dynamics.

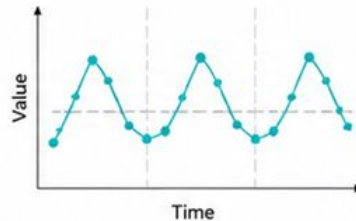


Example: Understand how river discharge changes throughout the years.



2. Detect Patterns

Identify recurring patterns such as seasonality, cycles, and regular fluctuations.

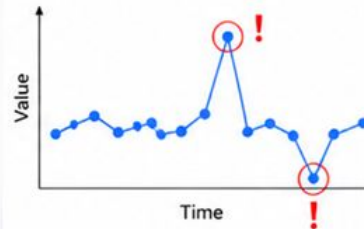


Example: Detect seasonal peaks in energy consumption every summer.



3. Detect Changes / Disturbances

Find structural breaks, anomalies, or unexpected events that indicate changes in the system.

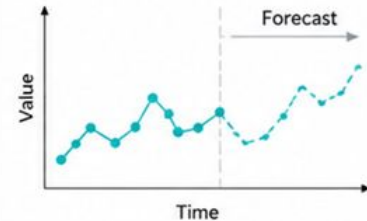


Example: Detect a flood event or a sensor failure in real time.



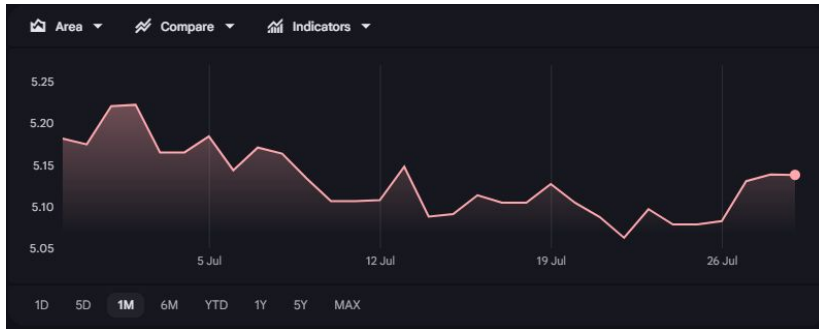
4. Perform Predictions

Use past information to forecast future values and support decision making.



Example: Forecast next month's demand to plan inventory.

Time Series



Time Series



Time Series



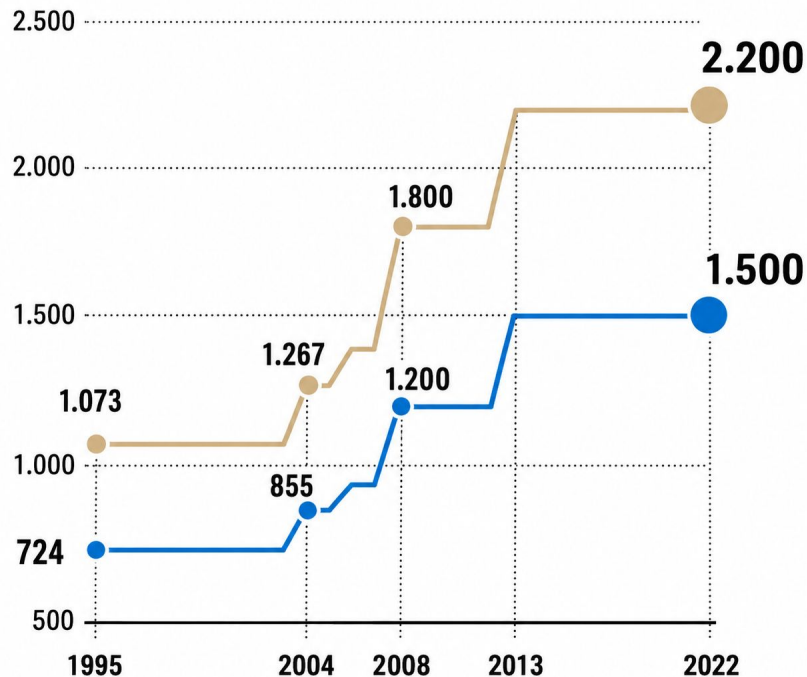
Time Series

Bolsas de estudos CAPES/CNPq

EM REAIS

MESTRADO

DOUTORADO



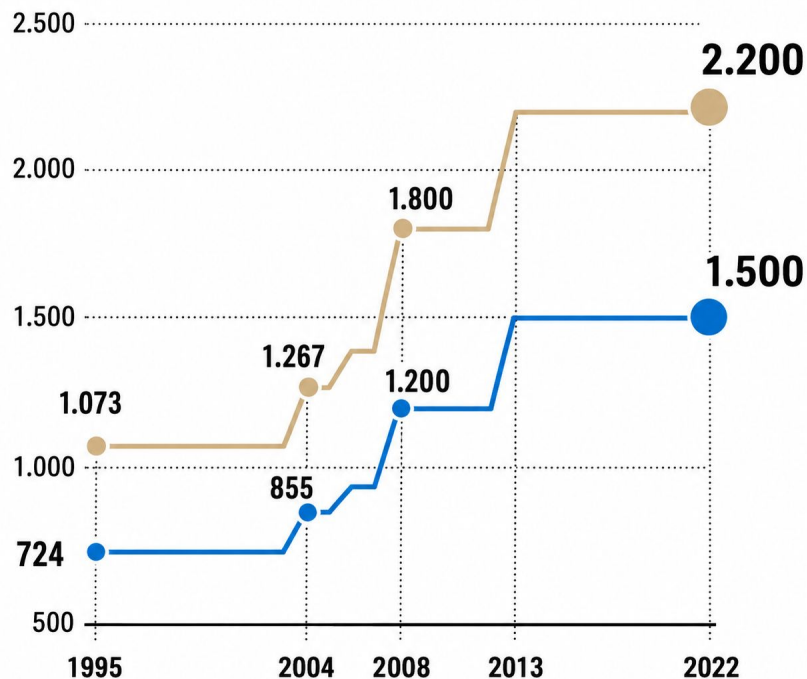
Time Series

Bolsas de estudos CAPES/CNPq

EM REAIS

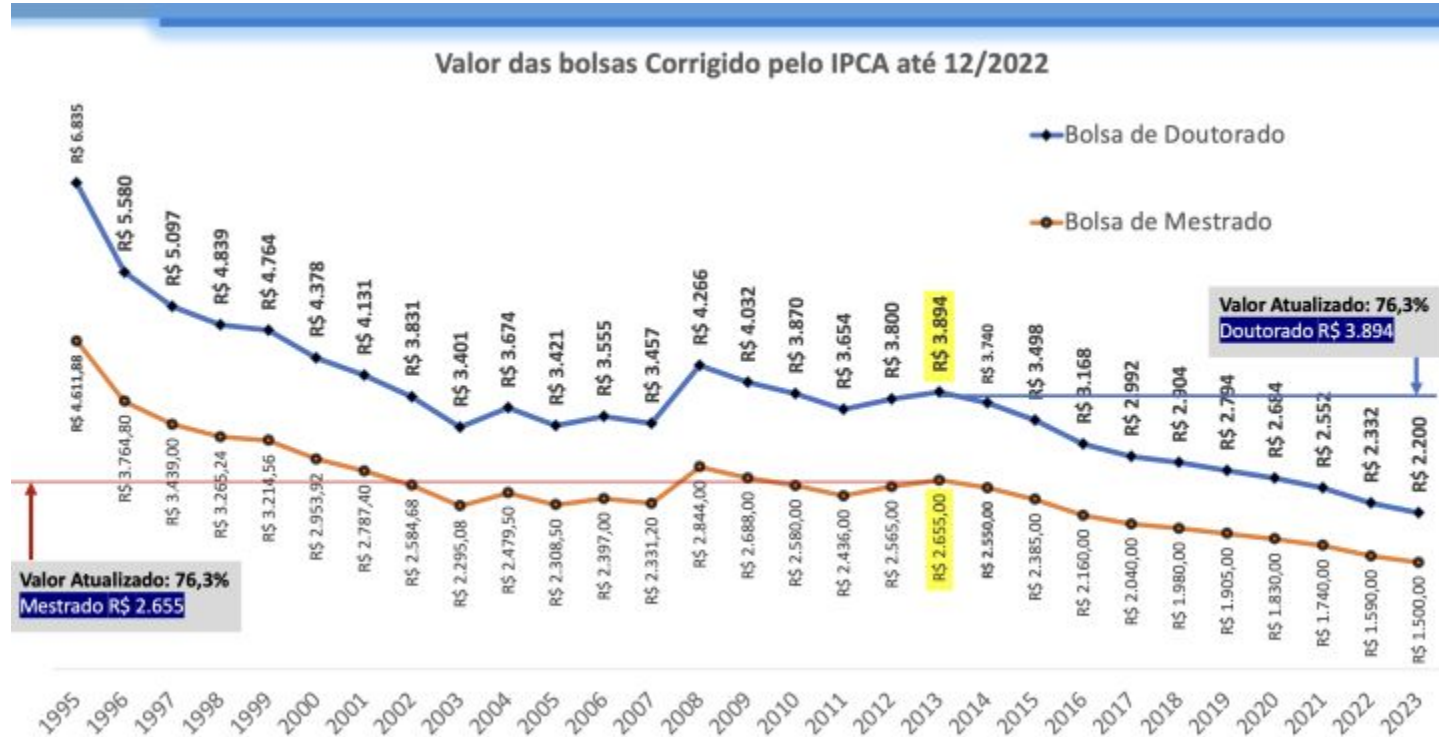
MESTRADO

DOUTORADO



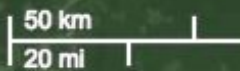
https://www.estadao.com.br/brasil/bolsas-da-capes-sem-reajuste-ha-9-anos-qual-seria-o-valor-do-auxilio-corrigido-pela-inflacao/?srsltid=AfmBOorVPvb08HfP6swBVsYbqPrhZS_I7mS_xsGU2t251012aQGOz9Pw

Time Series



Time Series

- Health
- Industry
- Shopping
- Climatology
- Meteorology
- Remote Sensing



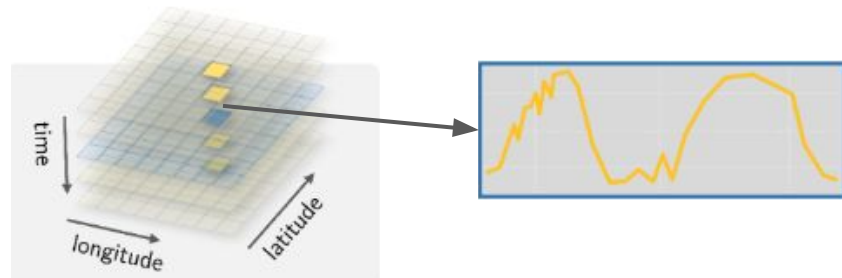
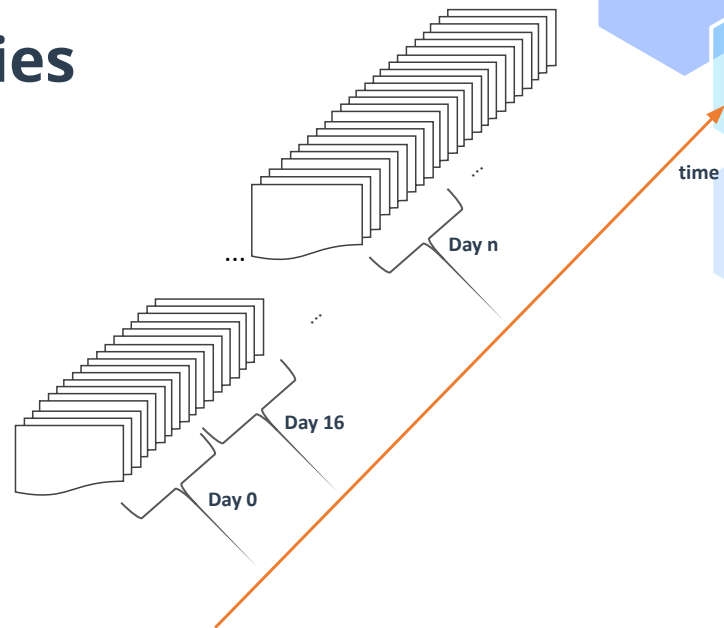
Fast

1984

Google

2012

Time Series

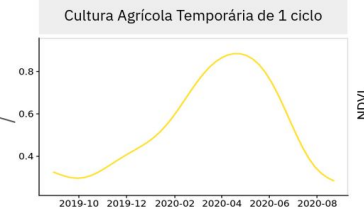
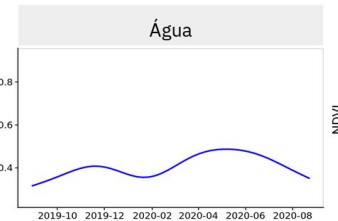
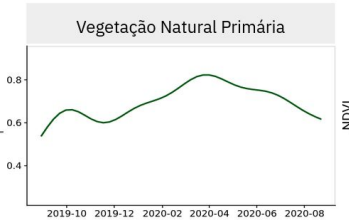
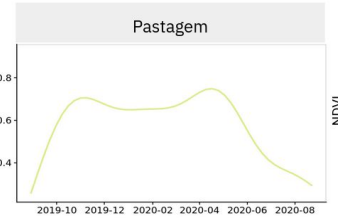
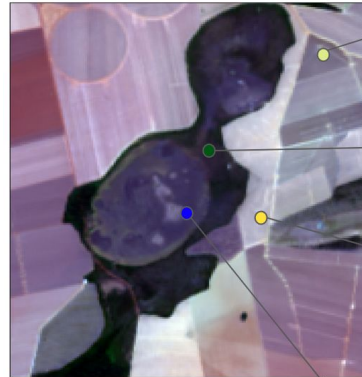


Time Series Characterization

Spectral indices (NDVI, EVI, NDWI) can show the behavior of fenology over the time;

Landsat has the longest time series

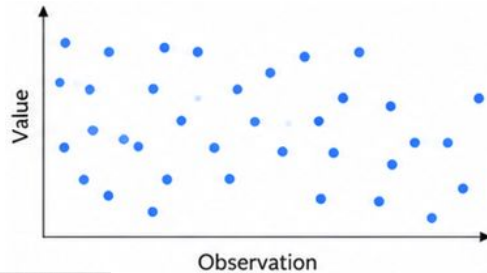
Each pixel is analysed over time



Time Series vs. i.i.d. data

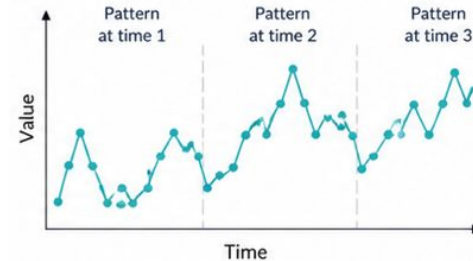
i.i.d. Data (Independent and identically distributed)

- Each observation is sampled independently of the others;
- All observations come from the same distribution;
- The order of data does not matter. You can shuffle the data without losing information.



Time series data

- Observations are ordered in time;
- Observations are dependent – today's value carries information about tomorrow's value (autocorrelation);
- The distribution can change over time (trend, seasonality, drift).



Time series data have structure in time. Ignoring that structure and treating it as i.i.d. data leads to wrong assumptions, misleading evaluations, and poor models.

Why this difference matters for ML

Treating time series as i.i.d. leads to wrong conclusions

What Happens

1



Traditional CV Breaks
(Data Leakage)

i.i.d. Data

Random K-fold or shuffle is valid.

Time Series Data

Random splits leak future information into training.

2



Statistical Assumptions Fail
(i.i.d. Assumption Violation)

i.i.d. Data

Many algorithms assume i.i.d. for theoretical guarantees.

Time Series Data

Observations are correlated. Residuals show autocorrelation, violating i.i.d. assumptions.

3



Features Must Capture Temporal Dependence
(Feature Engineering)

i.i.d. Data

Each row is self-contained. No need for temporal context.

Time Series Data

Need lags, rolling statistics, seasonality, and other temporal features to capture patterns.

4



Distribution Is Not Stationary
(Concept Drift)

i.i.d. Data

Assumes a fixed distribution over time.

Time Series Data

Distribution can change (trend, seasonality shift, structural breaks, drift).

Why It Matters for ML



Future information inflates performance. Models look better in validation but fail in real-world deployment. Use time-based splits or `TimeSeriesSplit`.



Impacts inference and reliability. Standard errors, confidence intervals, p-values, and some model assumptions become invalid.



Otherwise, the model misses the most valuable signal the temporal pattern.



Models may degrade over time. A model trained in 2020 may not work in 2024 without adaptation or retraining.



Components: Trend

Trend

Definition



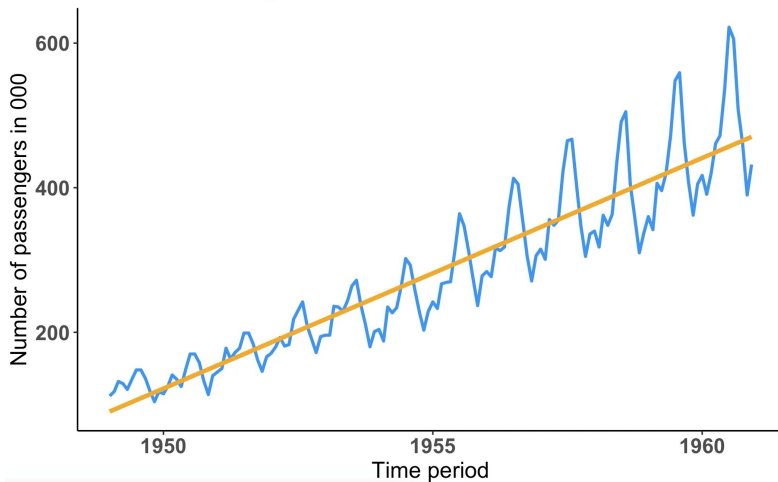
The long-term movement or direction of the series over time.

Key Points

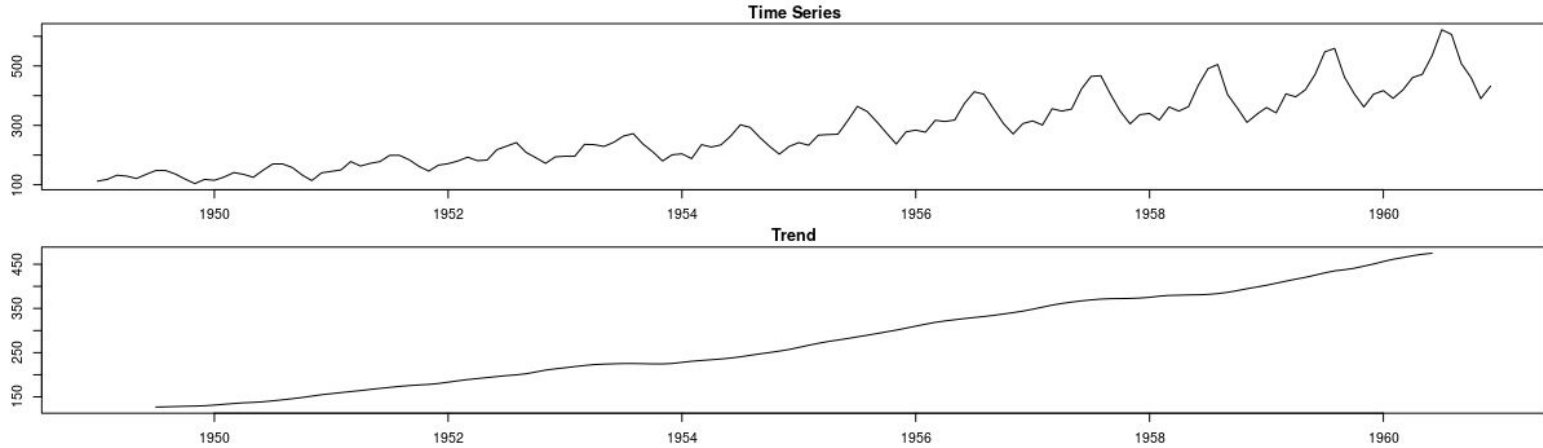
- Represents the general increase, decrease, or stability of the series.
- Occurs over long periods of time.
- Helps to understand the overall behavior of the data.

Example: Growth in population, climate change, or sales over the years.

Airline Passengers Dataset with a Trend Line



Time Series Components



```
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt

# Load dataset
data = sm.datasets.get_rdataset('AirPassengers').data
data['time'] = pd.date_range(start='1949-01', periods=len(data), freq='M')

# Decompose the time series multiplicatively
decomposition = sm.tsa.seasonal_decompose(data['value'], model='multiplicative', period=12)

# Plotting the decomposition
fig, (ax1, ax2, ax3, ax4) = plt.subplots(4, 1, figsize=(12, 8))
ax1.plot(data['value'], label='Original')
ax1.legend(loc='upper left')
ax2.plot(decomposition.trend, label='Trend')
ax2.legend(loc='upper left')
```

What is Time Series Trend in

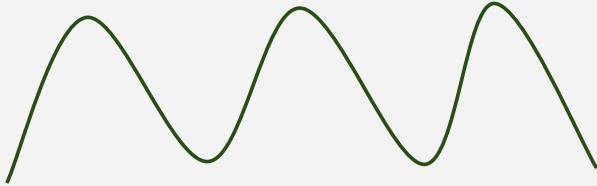
- Health
- Industry
- Shopping
- Climatology
- Meteorology
- Remote Sensing



Components: Seasonality

Seasonality

Pattern that repeats over fixed, known cycles



Definition

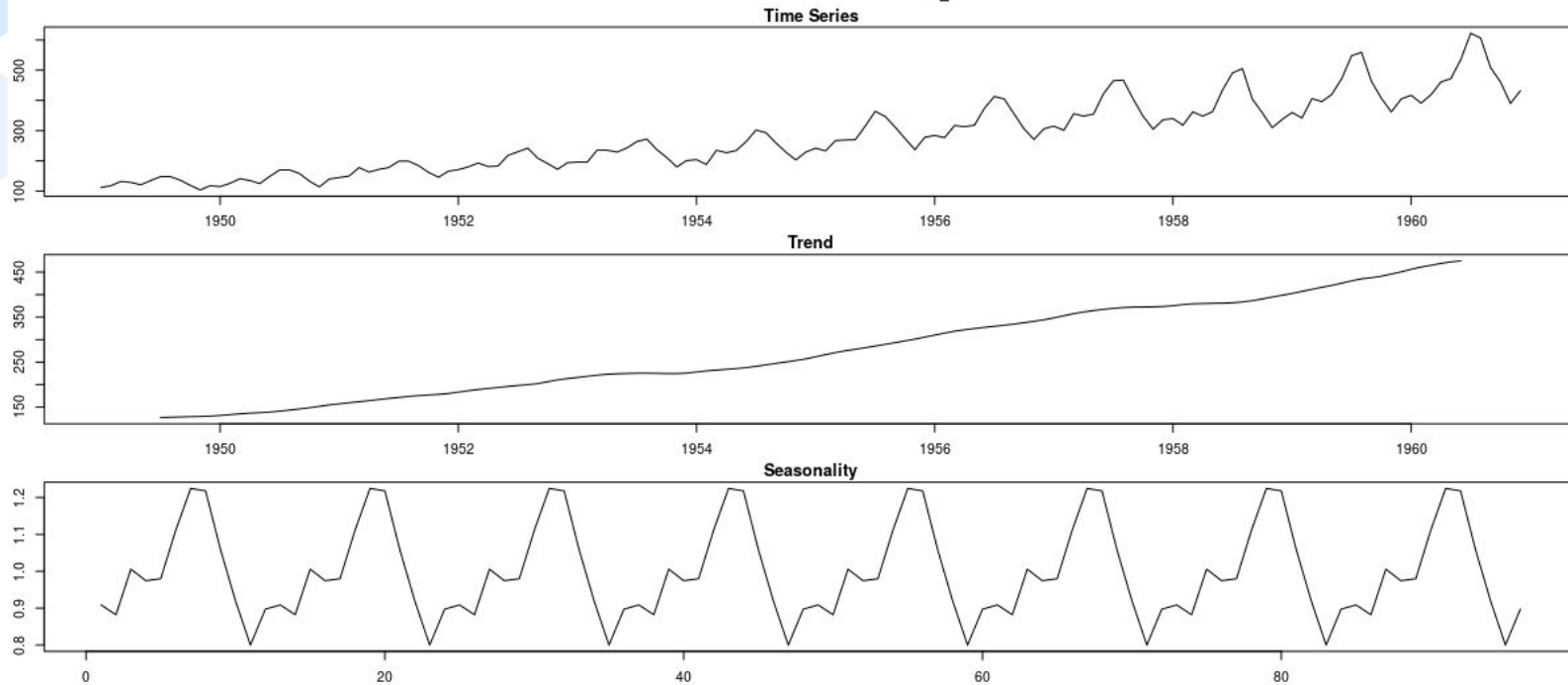
The regular and predictable pattern that repeats over a fixed period of time, such as daily, weekly, monthly, or yearly.

Key Points

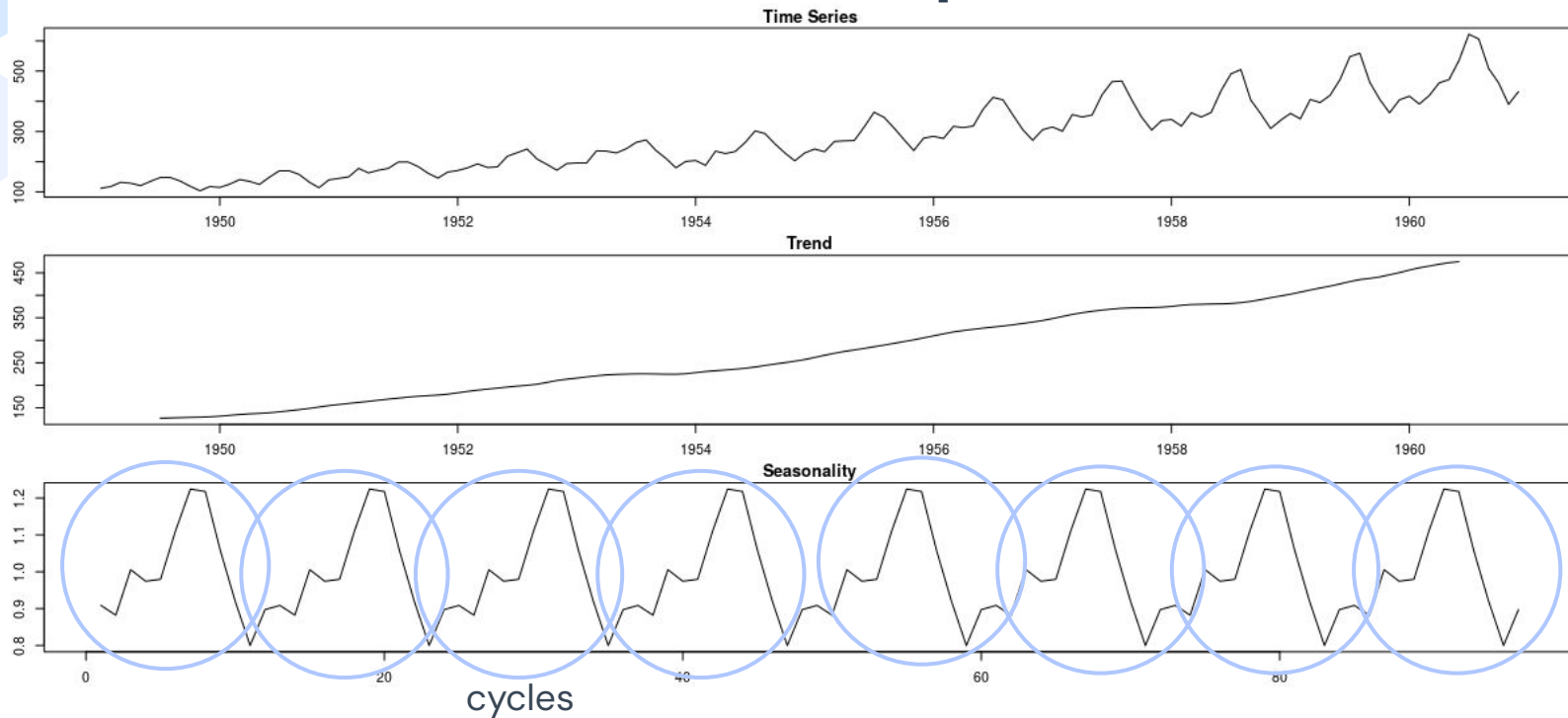
- Reflects systematic and repeating influences.
- Has a fixed period (e.g., 12 months, 7 days).
- Can be additive (constant amplitude) or multiplicative (changing amplitude).
- Important for understanding and forecasting future behavior.

Example: Higher energy consumption in summer, more sales in December, or rainfall peaks during a specific season.

Time Series Components



Time Series Components



What are Time Series Cycles in

- Health
- Industry
- Shopping
- Climatology
- Meteorology
- Remote Sensing



Components: Noise

Noise

Residual random variation not explained by the other components



Definition

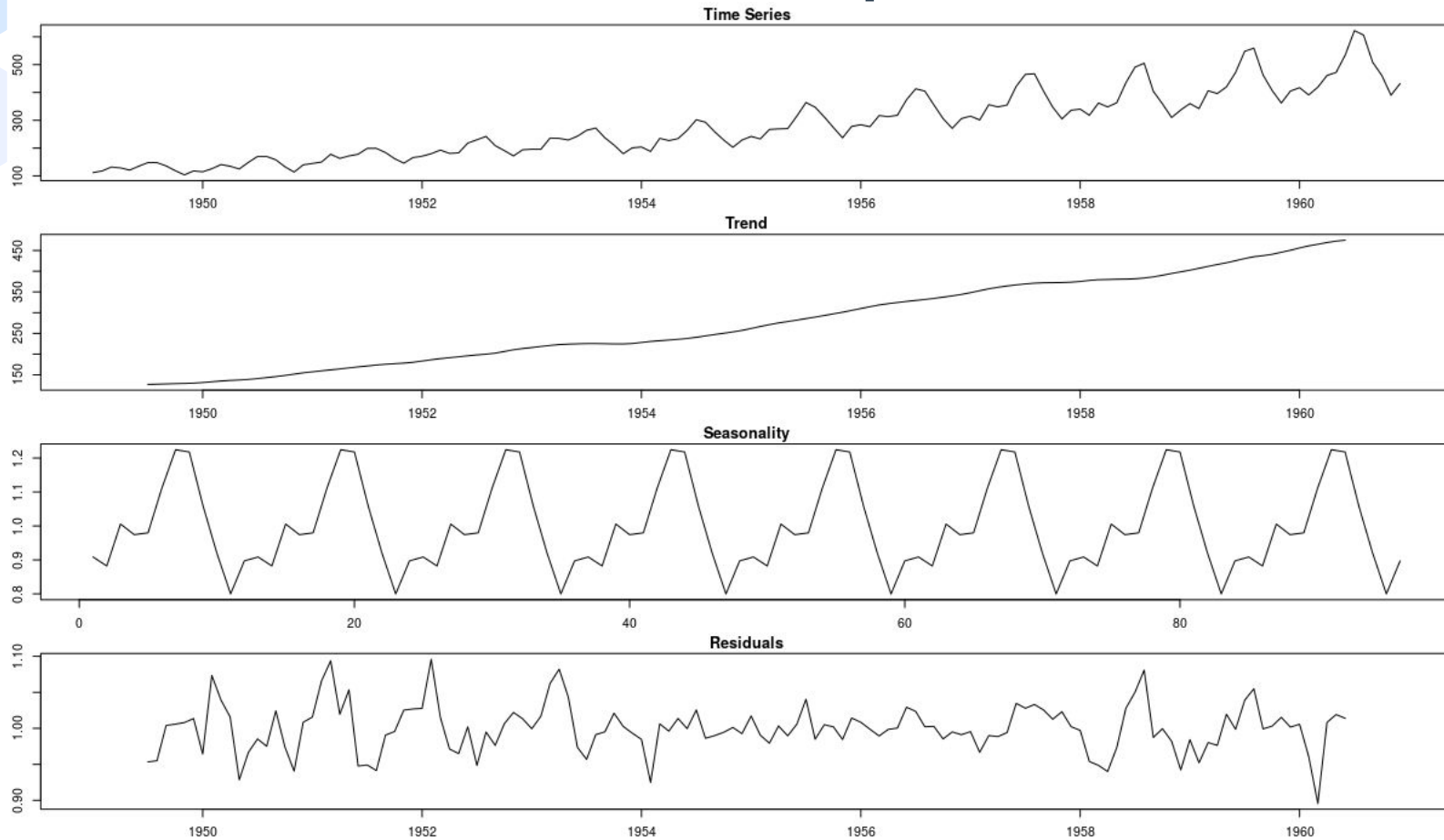
The irregular and unpredictable fluctuations in the series that cannot be explained by trend or seasonality.

Key Points

- Represents random or accidental influences.
- Has no fixed pattern or period.
- Caused by measurement errors, unusual events, or other unpredictable factors.
- Difficult to model, but important to consider in analysis.

Example: Sudden sensor errors, individual behavior changes, or unexpected external shocks (e.g., storms, strikes, promotions)..

Time Series Components



Time Series Components

```
import pandas as pd
import statsmodels.api as sm
import matplotlib.pyplot as plt
```

```
# Load dataset
```

```
data = sm.datasets.get_rdataset('AirPassengers').data
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                             periods=len(data), freq='M')
```

```
# Decompose the time series multiplicatively
decomposition =
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                           model='multiplicative', period=12)
```

```
# Plotting the decomposition
```

```
fig, (ax1, ax2, ax3, ax4) = plt.subplots(4, 1, figsize=(12, 8))
```

```
ax1.plot(data['value'], label='Original')
```

```
ax1.legend(loc='upper left')
```

```
ax2.plot(decomposition.trend, label='Trend')
```

```
ax2.legend(loc='upper left')
```

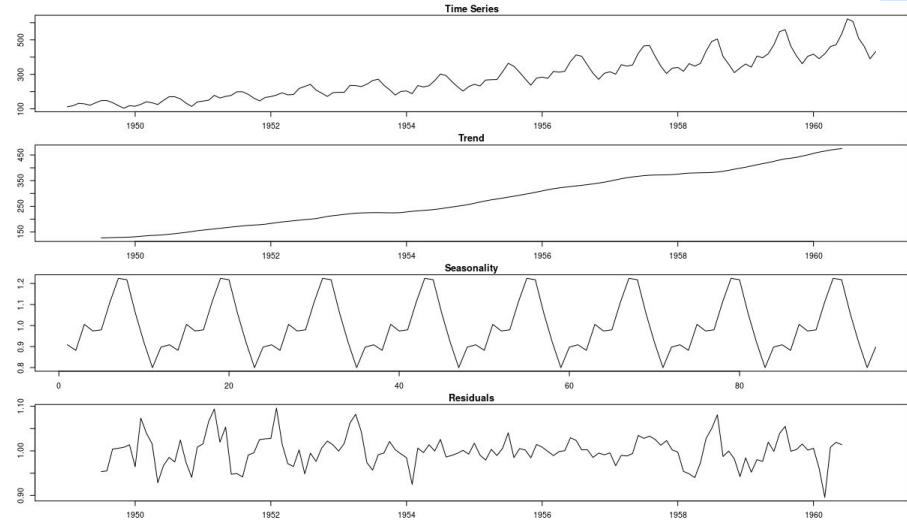
```
ax3.plot(decomposition.seasonal, label='Seasonal')
```

```
ax3.legend(loc='upper left')
```

```
ax4.plot(decomposition.resid, label='Residual')
```

```
ax4.legend(loc='upper left')
```

```
plt.show()
```



What is a Time Series Noise in

- Health
- Industry
- Shopping
- Climatology
- Meteorology
- Remote Sensing



Gaps and smoothing

Gaps and smoothing



Gaps and smoothing

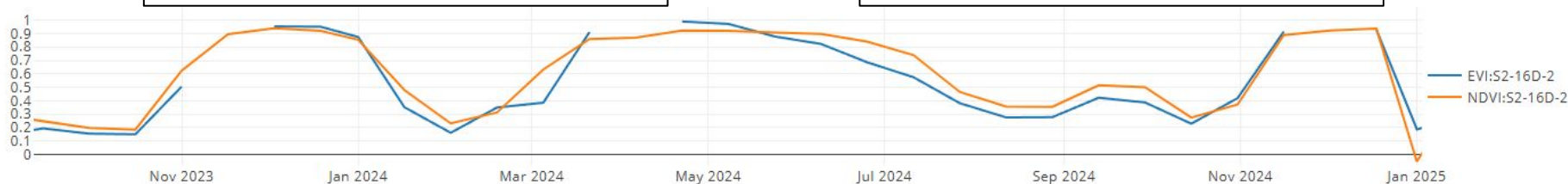
Simple substitution methods

- Substitution by zero
- Substitution by the mean
- Substitution by the previous or subsequent value

Interpolation

- Linear interpolation
- Polynomial interpolation
- Spline interpolation

Série Temporal



Statistical modeling

- Moving average models
- ARIMA models
- Exponential smoothing models

ML modeling

ML modeling

- Time-series-specific methods
- Imputation with decomposition
- Fourier-based imputation

Gaps and smoothing

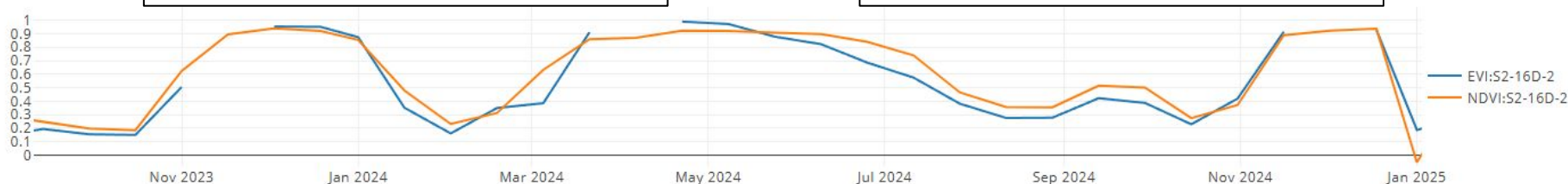
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Série Temporal



Statistical modeling

- Moving average models
- ARIMA models
- Exponential smoothing models

ML modeling

ML modeling

- Time-series-specific methods
- Imputation with decomposition
- Fourier-based imputation

**Is Linear Regression useful
to predict a time series?**



Remote Sensing Time Series

Time Series Characterization

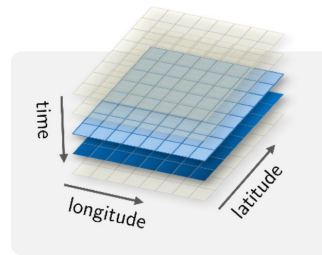
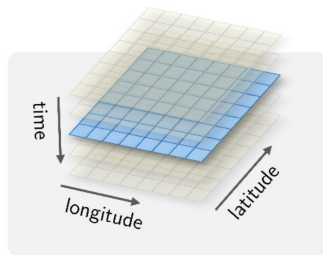
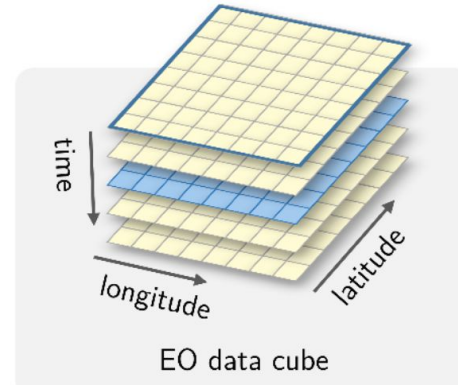
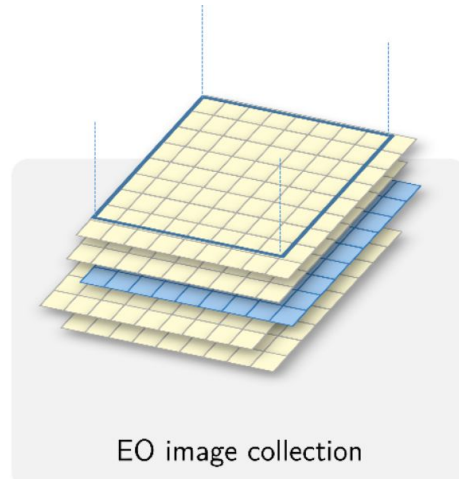
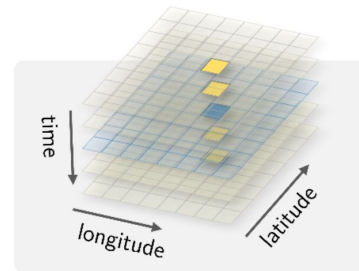


Image selection



Time series extraction

Data Cube Explorer: WLTS e WTSS

<https://data.inpe.br/bdc/explorer>



Remote Sensing Time Series Challenges

- Heterogeneous Spatial Resolution
- Harmonization
- Image Registration
- Cloud cover and shadows
- Irregular Revisit

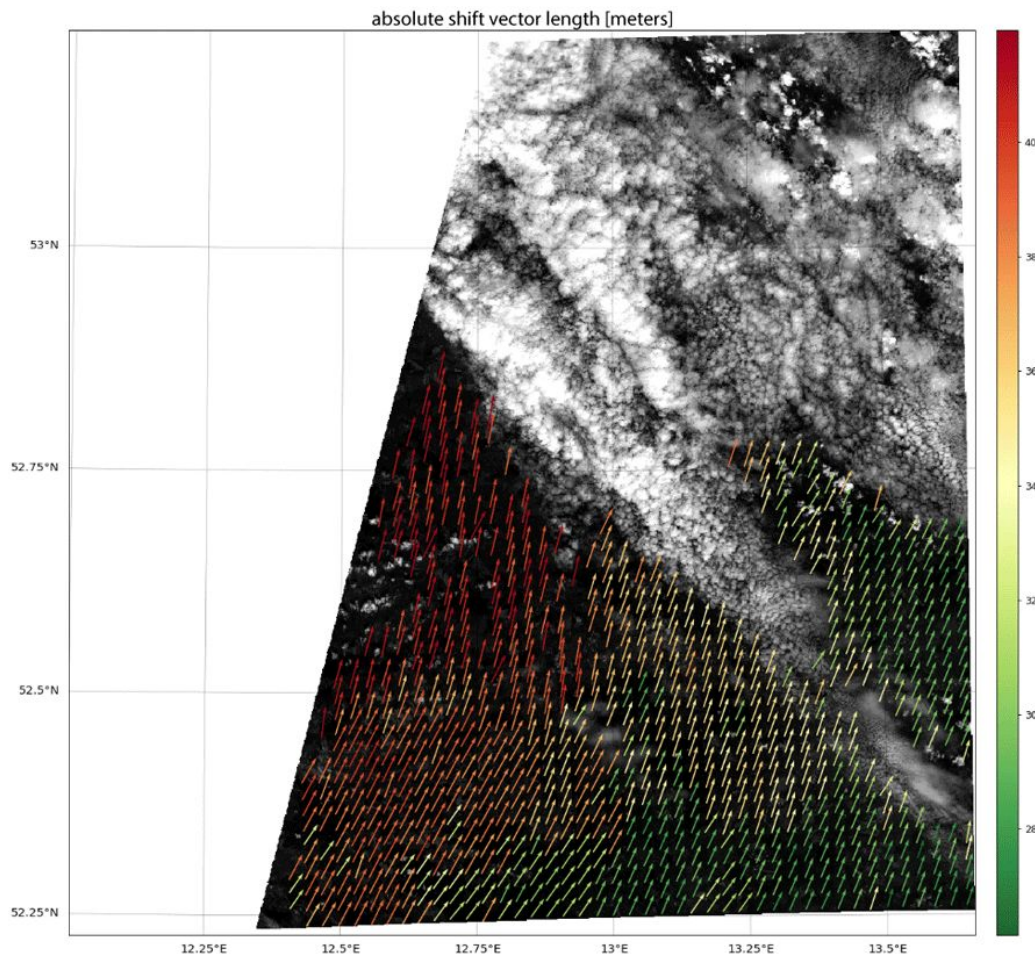


Remote Sensing Time Series challenges: Image Registration

Image registration



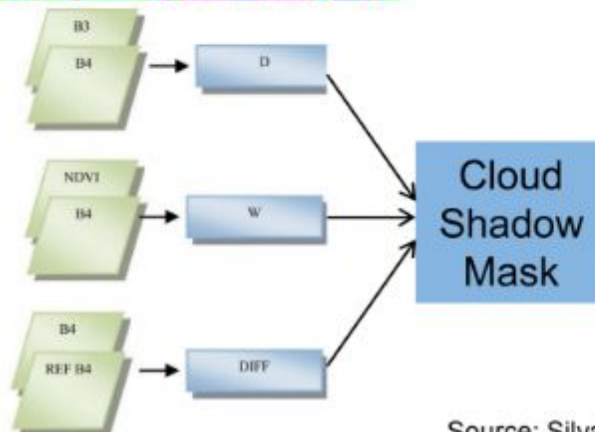
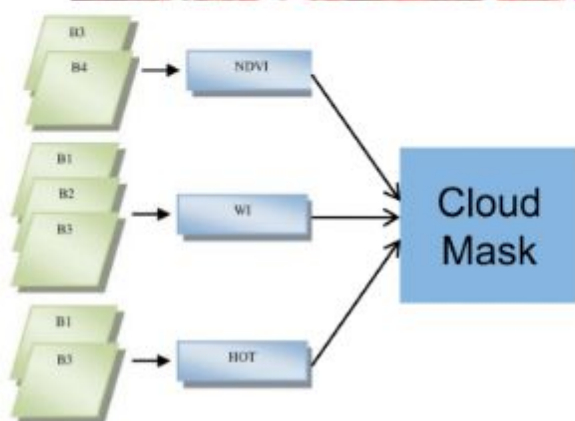
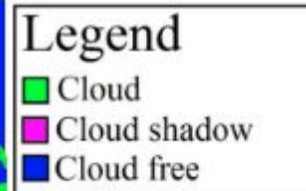
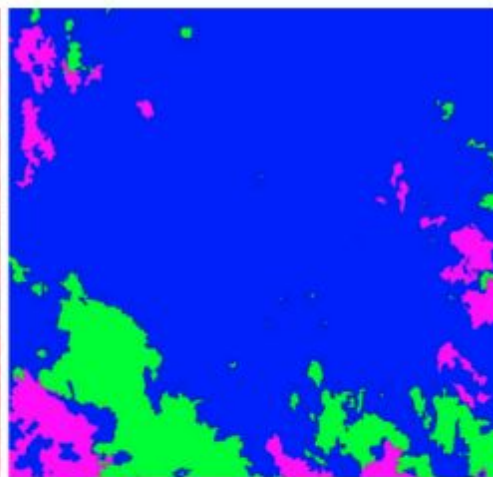
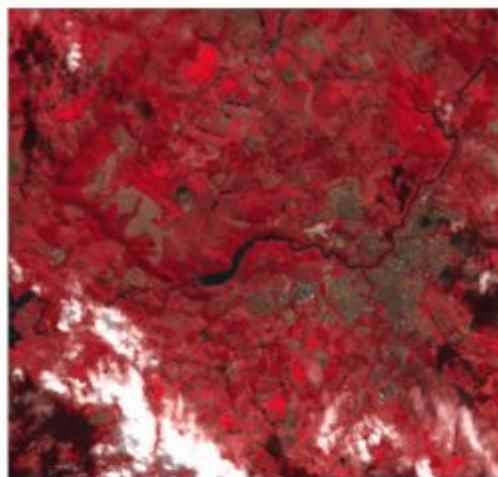
Image registration





Remote Sensing Time Series challenges: Cloud cover and shadows

CBERS Cloud Mask



Cloud Mask

Fmask 4

Clear	0 => clear land pixel
	1 => clear water pixel
Not Clear	2 => cloud shadow
	3 => snow
	4 => cloud
	255 => no observation

CMASK

	0 => nodata
Clear	127 => clear
Not Clear	255 => cloud

Sen2cor - SCL

Label	Classification
0	NO_DATA
1	SATURATED_OR_DEFECTIVE
2	DARK_AREA_PIXELS
3	CLOUD_SHADOWS
4	VEGETATION
5	NOT_VEGETATED
6	WATER
7	UNCLASSIFIED
8	CLOUD_MEDIUM_PROBABILITY
9	CLOUD_HIGH_PROBABILITY
10	THIN_CIRRUS
11	SNOW

Figure 3: Scene Classification Values

Landsat-8/9 C2 QA_PIXEL

Bit	Flag Description	Values
0	Fill	0 for image data 1 for fill data
1	Dilated Cloud	0 for cloud is not dilated or no cloud 1 for cloud dilation
2	Cirrus	0 for Cirrus Confidence: no confidence level set or Low Confidence 1 for high confidence cirrus
3	Cloud	0 for cloud confidence is not high 1 for high confidence cloud
4	Cloud Shadow	0 for Cloud Shadow Confidence is not high 1 for high confidence cloud shadow
5	Snow	0 for Snow/Ice Confidence is not high 1 for high confidence snow cover
6	Clear	0 if Cloud or Dilated Cloud bits are set 1 if Cloud and Dilated Cloud bits are not set
7	Water	0 for land or cloud 1 for water
8-9	Cloud Confidence	00 for no confidence level set 01 Low confidence 10 Medium confidence 11 High confidence
10-11	Cloud Shadow Confidence	00 for no confidence level set 01 Low confidence 10 Reserved 11 High confidence
12-13	Snow/Ice Confidence	00 for no confidence level set 01 Low confidence 10 Reserved 11 High confidence
14-15	Cirrus Confidence	00 for no confidence level set 01 Low confidence 10 Reserved 11 High confidence

Table 6-2. Landsat 8 Pixel Quality Assessment (QA_PIXEL) Bit Index

Pixel Value	Fill	Dilated Cloud	Cirrus	Cloud	Cloud Shadow	Snow	Clear	Water	Cloud Conf.	Cloud Shadow Conf.	Snow/Ice Conf.	Cirrus Conf.	Pixel Description
1	Yes	No	No	No	No	No	No	No	None	None	None	None	Fill
21824	No	No	No	No	No	No	Yes	No	Low	Low	Low	Low	Clear with lows set
21826	No	Yes	No	No	No	No	Yes	No	Low	Low	Low	Low	Dilated cloud over land
21888	No	No	No	No	No	No	No	Yes	Low	Low	Low	Low	Water with lows set
21890	No	Yes	No	No	No	No	No	Yes	Low	Low	Low	Low	Dilated cloud over water
22080	No	No	No	No	No	No	Yes	No	Mid	Low	Low	Low	Mid conf cloud
22144	No	No	No	No	No	No	No	Yes	Mid	Low	Low	Low	Mid conf cloud over water
22280	No	No	No	Yes	No	No	No	No	High	Low	Low	Low	High conf Cloud
23888	No	No	No	No	Yes	No	Yes	No	Low	High	Low	Low	High conf cloud shadow
23952	No	No	No	No	Yes	No	No	Yes	Low	High	Low	Low	Water with cloud shadow
24088	No	No	No	Yes	Yes	No	No	No	Mid	High	Low	Low	Mid conf cloud with shadow
24216	No	No	No	Yes	Yes	No	No	Yes	Mid	High	Low	Low	Mid conf cloud with shadow over water
24344	No	No	No	Yes	Yes	No	No	No	High	High	Low	Low	High conf cloud with shadow
24472	No	No	No	Yes	Yes	No	No	Yes	High	High	Low	Low	High conf cloud with shadow over water
30048	No	No	No	No	No	Yes	Yes	No	Low	Low	High	Low	High conf snow/ice
54596	No	No	Yes	No	No	No	Yes	No	Low	Low	Low	High	High conf Cirrus
54852	No	No	Yes	No	No	No	Yes	No	Mid	Low	Low	High	Cirrus, mid cloud
55052	No	No	Yes	Yes	No	No	No	No	High	Low	Low	High	Cirrus, high cloud
56856	No	No	No	Yes	Yes	No	No	No	Mid	High	Low	High	Cirrus, mid conf cloud, shadow
56984	No	No	No	Yes	Yes	No	No	Yes	Mid	High	Low	High	Cirrus, mid conf cloud, shadow, over water
57240	No	No	No	Yes	Yes	No	No	Yes	High	High	Low	High	Cirrus, high conf cloud, shadow

Table 6-3. Landsat 8 Pixel Quality Assessment (QA_PIXEL) Value Interpretations

Landsat-8/9 C2 QA_PIXEL

24216

0101 1110 1001 1000

Bit	Flag Description	Values
0	Fill	0 for image data 1 for fill data
1	Dilated Cloud	0 for cloud is not dilated or no cloud 1 for cloud dilation
2	Cirrus	0 for Cirrus Confidence: no confidence level set or Low Confidence 1 for high confidence cirrus
3	Cloud	0 for cloud confidence is not high 1 for high confidence cloud
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5	Snow	0 for Snow/Ice Confidence is not high 1 for high confidence snow cover
6	Clear	0 if Cloud or Dilated Cloud bits are set 1 if Cloud and Dilated Cloud bits are not set
7	Water	0 for land or cloud 1 for water
8-9	Cloud Confidence	00 for no confidence level set
		01 Low confidence
		10 Medium confidence
		11 High confidence
10-11	Cloud Shadow Confidence	00 for no confidence level set
		01 Low confidence
		10 Reserved
		11 High confidence
12-13	Snow/Ice Confidence	00 for no confidence level set
		01 Low confidence
		10 Reserved
		11 High confidence
14-15	Cirrus Confidence	00 for no confidence level set
		01 Low confidence
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1	Yes	No	No	No	No	No	No	No	None	None	None	None	Fill
21824	No	No	No	No	No	No	Yes	No	Low	Low	Low	Low	Clear with lows set
21826	No	Yes	No	No	No	No	Yes	No	Low	Low	Low	Low	Dilated cloud over land
21888	No	No	No	No	No	No	No	Yes	Low	Low	Low	Low	Water with lows set
21890	No	Yes	No	No	No	No	No	Yes	Low	Low	Low	Low	Dilated cloud over water
22080	No	No	No	No	No	No	Yes	No	Mid	Low	Low	Low	Mid conf cloud
22144	No	No	No	No	No	No	No	Yes	Mid	Low	Low	Low	Mid conf cloud over water
22280	No	No	No	Yes	No	No	No	No	High	Low	Low	Low	High conf Cloud
23888	No	No	No	No	Yes	No	Yes	No	Low	High	Low	Low	High conf cloud shadow
23952	...	...	...	...	...	...	...	...	...	...	...	...	Water with cloud shadow
24088	0	0	0	1	1	0	0	1	10	11	01	01	Mid conf cloud with shadow
24216	No	No	No	Yes	Yes	No	No	Yes	Mid	High	Low	Low	Mid conf cloud with shadow over water
24344	No	No	No	Yes	Yes	No	No	No	High	High	Low	Low	High conf cloud with shadow
24472	No	No	No	Yes	Yes	No	No	Yes	High	High	Low	Low	High conf cloud with shadow over water
30048	No	No	No	No	No	Yes	Yes	No	Low	Low	High	Low	High conf snow/ice
54596	No	No	Yes	No	No	No	Yes	No	Low	Low	Low	High	High conf Cirrus
54852	No	No	Yes	No	No	No	Yes	No	Mid	Low	Low	High	Cirrus, mid cloud
55052	No	No	Yes	Yes	No	No	No	No	High	Low	Low	High	Cirrus, high cloud
56856	No	No	No	Yes	Yes	No	No	No	Mid	High	Low	High	Cirrus, mid conf cloud, shadow
56984	No	No	No	Yes	Yes	No	No	Yes	Mid	High	Low	High	Cirrus, mid conf cloud, shadow, over water
57240	No	No	No	Yes	Yes	No	No	Yes	High	High	Low	High	Cirrus, high conf cloud, shadow

Table 6-3. Landsat 8 Pixel Quality Assessment (QA_PIXEL) Value Interpretations

Landsat-8/9 C2 QA_PIXEL

Landsat 8 OLI/TIRS Collection 1 Level-1 Quality Assessment Band Attribute and Pixel Value Possibilities	
Attribute	Pixel Value
Fill	1
Terrain Occlusion	2, 2722
Clear	2720, 2724, 2728, 2732
Radiometric Saturation - 1-2 bands	2724, 2756, 2804, 2980, 3012, 3748, 3780, 6820, 6852, 6900, 7076, 7108, 7844, 7876
Radiometric Saturation - 3-4 bands	2728, 2760, 2808, 2984, 3016, 3752, 3784, 6824, 6856, 6904, 7080, 7112, 7848, 7880
Radiometric Saturation - 5+ bands	2732, 2764, 2812, 2988, 3020, 3756, 3788, 6828, 6860, 6908, 7084, 7116, 7852, 7884
Cloud	2800, 2804, 2808, 2812, 6896, 6900, 6904, 6908
Cloud Confidence - Low	2720, 2722, 2724, 2728, 2732, 2976, 2980, 2984, 2988, 3744, 3748, 3752, 3756, 6816, 6820, 6824, 6828, 7072, 7076, 7080, 7084, 7840, 7844, 7848, 7852
Cloud Confidence - Medium	2752, 2756, 2760, 2764, 3008, 3012, 3016, 3020, 3776, 3780, 3784, 3788, 6848, 6852, 6856, 6860, 7104, 7108, 7112, 7116, 7872, 7876, 7880, 7884
Cloud Confidence - High	2800, 2804, 2808, 2812, 6896, 6900, 6904, 6908
Cloud Shadow - High	2976, 2980, 2984, 2988, 3008, 3012, 3016, 3020, 7072, 7076, 7080, 7084, 7104, 7108, 7112, 7116
Snow/Ice - High	3744, 3748, 3752, 3756, 3776, 3780, 3784, 3788, 7840, 7844, 7848, 7852, 7872, 7876, 7880, 7884
Cirrus Confidence - Low	2720, 2722, 2724, 2728, 2732, 2752, 2756, 2760, 2764, 2800, 2804, 2808, 2812, 2976, 2980, 2984, 2988, 3008, 3012, 3016, 3020, 3744, 3748, 3752, 3756, 3780, 3784, 3788
Cirrus Confidence - High	6816, 6820, 6824, 6828, 6848, 6852, 6856, 6860, 6896, 6900, 6904, 6908, 7072, 7076, 7080, 7084, 7104, 7108, 7112, 7116, 7840, 7844, 7848, 7852, 7872, 7876, 7880, 7884

Landsat-8/9 C2 QA_PIXEL

Band Designation	File Format	Band Name	Data Type	Units	Data Range	Valid Range	Fill Value	Multiplicative Scale Factor	Additive Scale Factor
ProductID_SR_B1	TIF	Band 1 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B2	TIF	Band 2 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B3	TIF	Band 3 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B4	TIF	Band 4 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B5	TIF	Band 5 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B6	TIF	Band 6 SR	UINT16	Reflectance	1 - 65535	1 - 65535	0	0.0000275	-0.2
ProductID_SR_B7	TIF	Band 7 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_ST_B10	TIF	Band 10 ST	UINT16	Kelvin	1 - 65535	1 - 65535	0	0.00341802	149
ProductID_QA_PIXEL	TIF	Level 2 Pixel Quality Band	UINT16	Bit Index	1 - 65535	21824 - 65534	1	NA	NA
ProductID_SR_QA_AEROSOL	TIF	Aerosol QA	UINT8	Bit Index	0 - 255	0 - 255	1	NA	NA
ProductID_QA_RADSAT	TIF	Radiometric Saturation QA	UINT16	Bit Index	0 - 32768	0 - 3829	NA	NA	NA
ProductID_ST_QA	TIF	Surface Temperature Uncertainty	INT16	Kelvin	0 - 32767	0 - 32767	-9999	0.01	NA
ProductID_ST_TRAD	TIF	Thermal Band in Radiance	UINT16	W/(m ² .sr.μm)/DN	0 - 22000	0 - 22000	-9999	0.001	NA
ProductID_ST_URAD	TIF	Upwelled Radiance	UINT16	W/(m ² .sr.μm)/DN	0 - 28000	0 - 28000	-9999	0.001	NA
ProductID_ST_DRAD	TIF	Downwelled Radiance	UINT16	W/(m ² .sr.μm)/DN	0 - 28000	0 - 28000	-9999	0.001	NA
ProductID_ST_ATRAN	TIF	Atmospheric Transmittance	UINT16	Unitless	0 - 10000	0 - 10000	-9999	0.0001	NA
ProductID_ST_EMIS	TIF	Emissivity estimated from ASTER GED	UINT16	Unitless	0 - 10000	0 - 10000	-9999	0.0001	NA
ProductID_ST_EMSD	TIF	Emissivity standard deviation	UINT16	Unitless	0 - 65535	0 - 10000	-9999	0.0001	NA
ProductID_ST_CDIST	TIF	Pixel distance to cloud	UINT16	Kilometers	0 - 24000	0 - 24000	-9999	0.01	NA
ProductID_MTL	txt and xml	Level 1 Metadata file	NA	NA	NA	NA	NA	NA	NA

Table 6-1. Collection 2 (C2) Level 2 Science Products (L2SP) Band Specifications

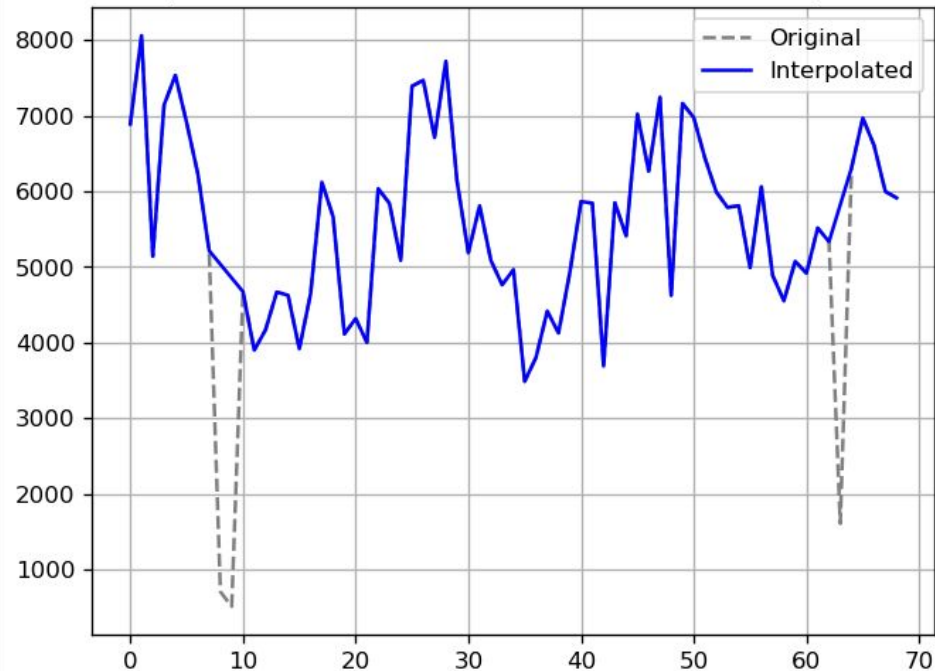
Landsat-8/9 C2 QA_PIXEL

Band Designation	File Format	Band Name	Data Type	Units	Data Range	Valid Range	Fill Value	Multiplicative Scale Factor	Additive Scale Factor
ProductID_SR_B1	TIF	Band 1 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B2	TIF	Band 2 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B3	TIF	Band 3 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B4	TIF	Band 4 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B5	TIF	Band 5 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B6	TIF	Band 6 SR	UINT16	Reflectance	1 - 65535	1 - 65535	0	0.0000275	-0.2
ProductID_SR_B7	TIF	Band 7 SR	UINT16	Reflectance	1 - 65455	7273 - 43636	0	0.0000275	-0.2
ProductID_ST_B10	TIF	Band 10 ST	UINT16	Kelvin	1 - 65535	1 - 65535	0	0.00341802	149
ProductID_QA_PIXEL	TIF	Level 2 Pixel Quality Band	UINT16	Bit Index	1 - 65535	21824 - 65534	1	NA	NA
ProductID_SR_QA_AEROSOL	TIF	Aerosol QA	UINT8	Bit Index	0 - 255	0 - 255	1	NA	NA
ProductID_QA_RADSAT	TIF	Radiometric Saturation QA	UINT16	Bit Index	0 - 32768	0 - 3829	NA	NA	NA
ProductID_ST_QA	TIF	Surface Temperature Uncertainty	INT16	Kelvin	0 - 32767	0 - 32767	-9999	0.01	NA
ProductID_ST_TRAD	TIF	Thermal Band in Radiance	UINT16	W/(m ² .sr.μm)/DN	0 - 22000	0 - 22000	-9999	0.001	NA
ProductID_ST_URAD	TIF	Upwelled Radiance	UINT16	W/(m ² .sr.μm)/DN	0 - 28000	0 - 28000	-9999	0.001	NA
ProductID_ST_DRAD	TIF	Downwelled Radiance	UINT16	W/(m ² .sr.μm)/DN	0 - 28000	0 - 28000	-9999	0.001	NA
ProductID_ST_ATRAN	TIF	Atmospheric Transmittance	UINT16	Unitless	0 - 10000	0 - 10000	-9999	0.0001	NA
ProductID_ST_EMIS	TIF	Emissivity estimated from ASTER GED	UINT16	Unitless	0 - 10000	0 - 10000	-9999	0.0001	NA
ProductID_ST_EMSD	TIF	Emissivity standard deviation	UINT16	Unitless	0 - 65535	0 - 10000	-9999	0.0001	NA
ProductID_ST_CDIST	TIF	Pixel distance to cloud	UINT16	Kilometers	0 - 24000	0 - 24000	-9999	0.01	NA
ProductID_MTL	txt and xml	Level 1 Metadata file	NA	NA	NA	NA	NA	NA	NA

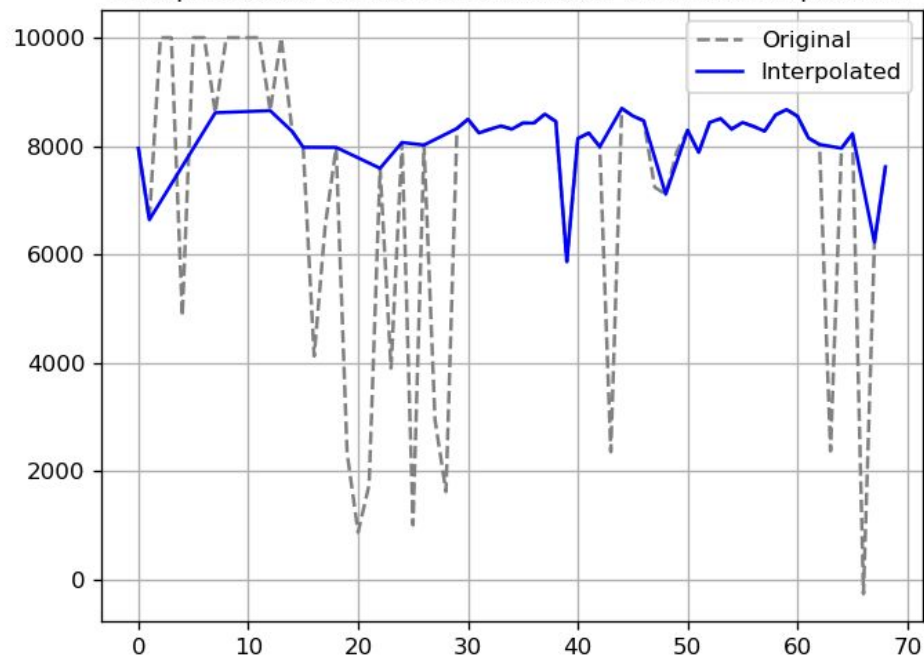
Table 6-1. Collection 2 (C2) Level 2 Science Products (L2SP) Band Specifications

Removing clouds from Time Series

Comparison of Time Series with and without interpolation



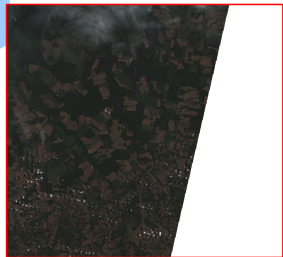
Comparison of Time Series with and without interpolation



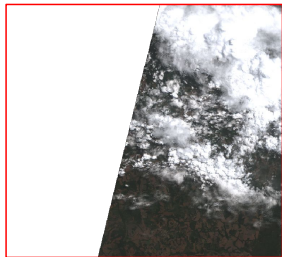


Remote Sensing Time Series challenges: Irregular revisit (equidistance)

Coleção original vs Cubos de Dados



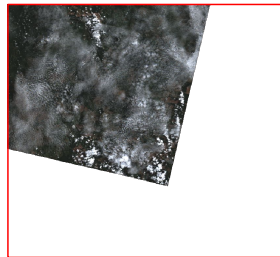
2025-03-18



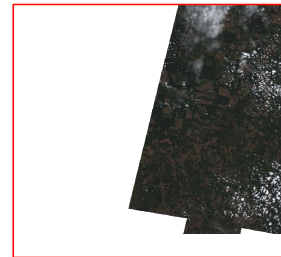
2025-03-20



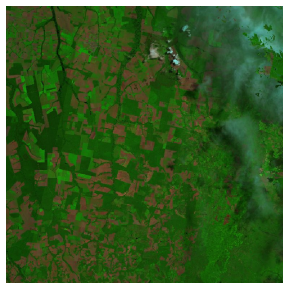
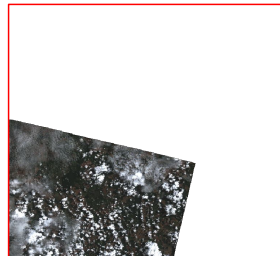
2025-03-22



2025-03-23

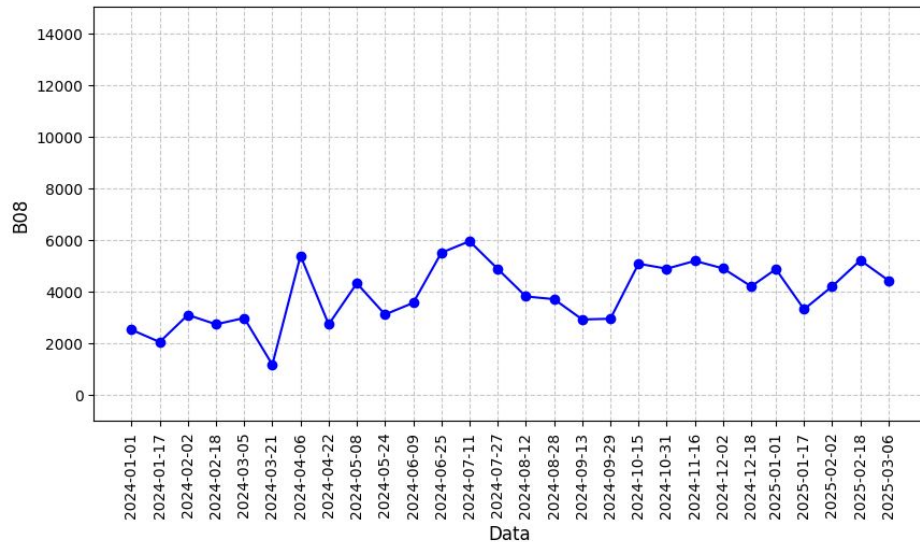
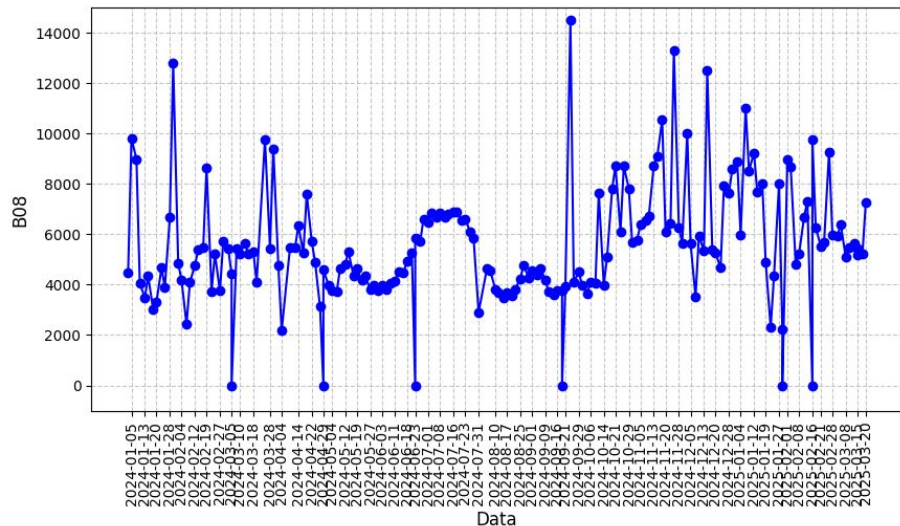


2025-03-25



16 dias “agrupados”

Original Collection vs Data Cubes



Missing data in time series

Imputation: four main approaches

1 Simple Fill Methods



Use nearby observed values to fill gaps with minimal assumptions.

Examples

- Remove (listwise deletion)
- Forward fill (last observation carried forward)
- Backward fill (next observation carried backward)
- Mean (global mean or window mean)

2 Interpolation Methods



Estimate missing values using mathematical functions between known points.

Examples

- Linear interpolation
- Polynomial interpolation
- Spline interpolation
- Time-weighted interpolation

3 Seasonal/Pattern-Aware Methods



Leverage seasonality or repeating patterns in the series to inform imputation.

Examples

- Seasonal decomposition + fill
- Same-period substitution (e.g., same day last year)
- Moving average with seasonal window
- STL or ETS-based filling

4 Model-Based Methods



Use statistical or machine learning models to predict missing values based on learned patterns.

Examples

- KNN imputation
- Regression models
- State space models (e.g., Kalman filter)
- Deep learning models (e.g., LSTM)

Most Simple

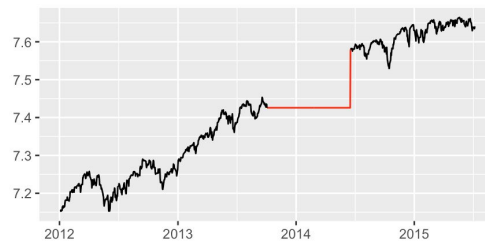
Most Sophisticated

Not equally resampled time series

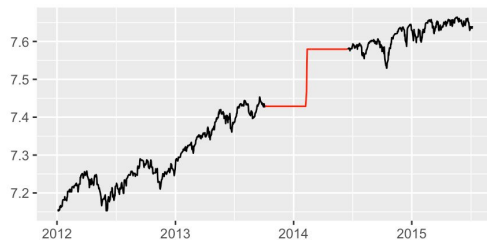
Original



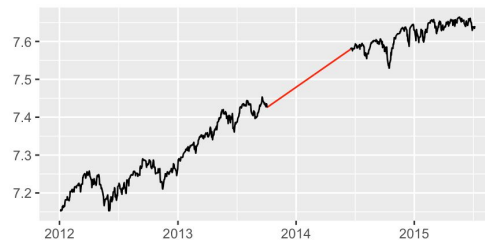
Imputation with LOCF



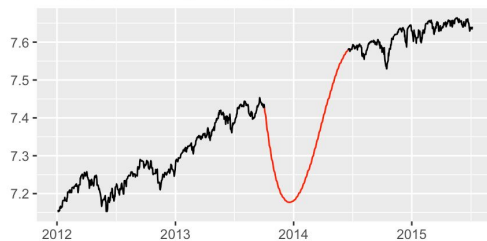
Imputation with MA



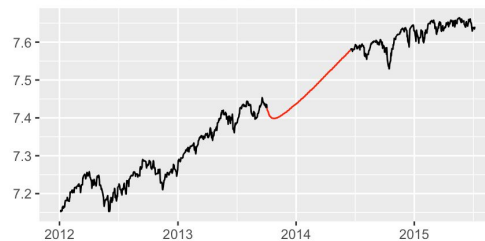
Imputation with linear interpolation



Imputation with spline interpolation



Imputation with Stineman interpolation

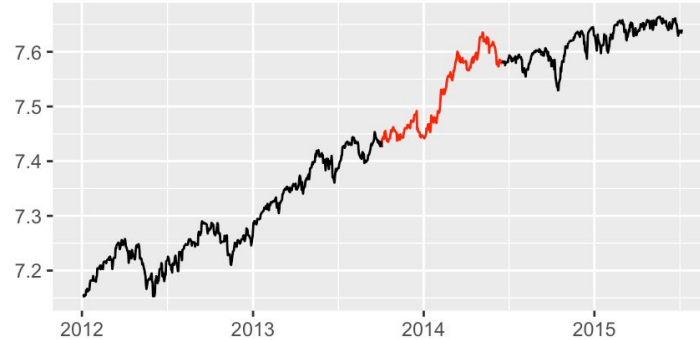


Not equally resampled time series

Original



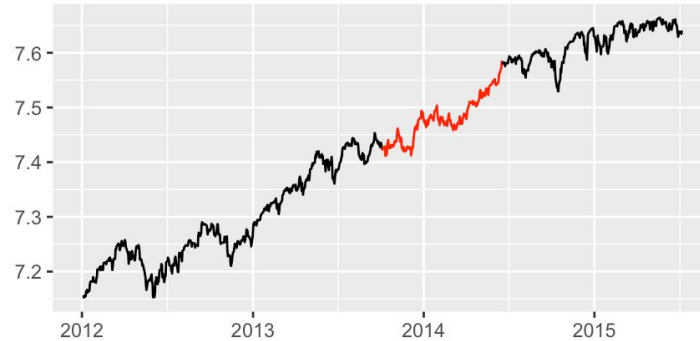
Imputation 1



Imputation 2



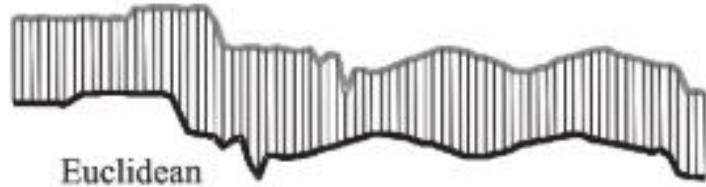
Imputation 3



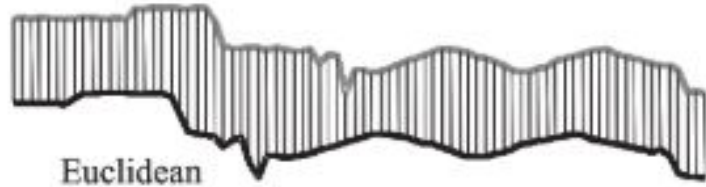


Time Series Comparison

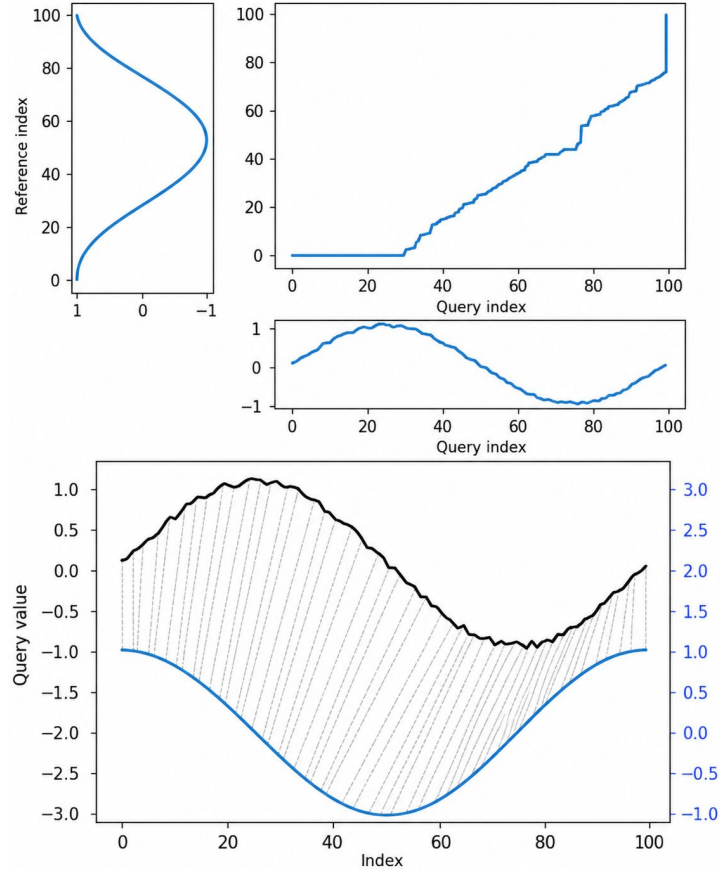
Time Series Comparison



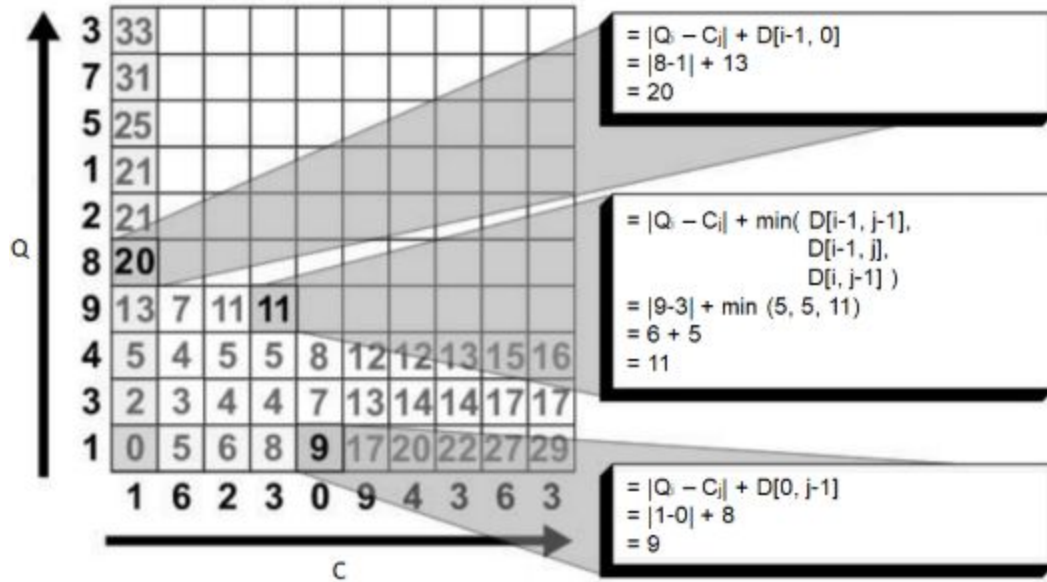
DTW



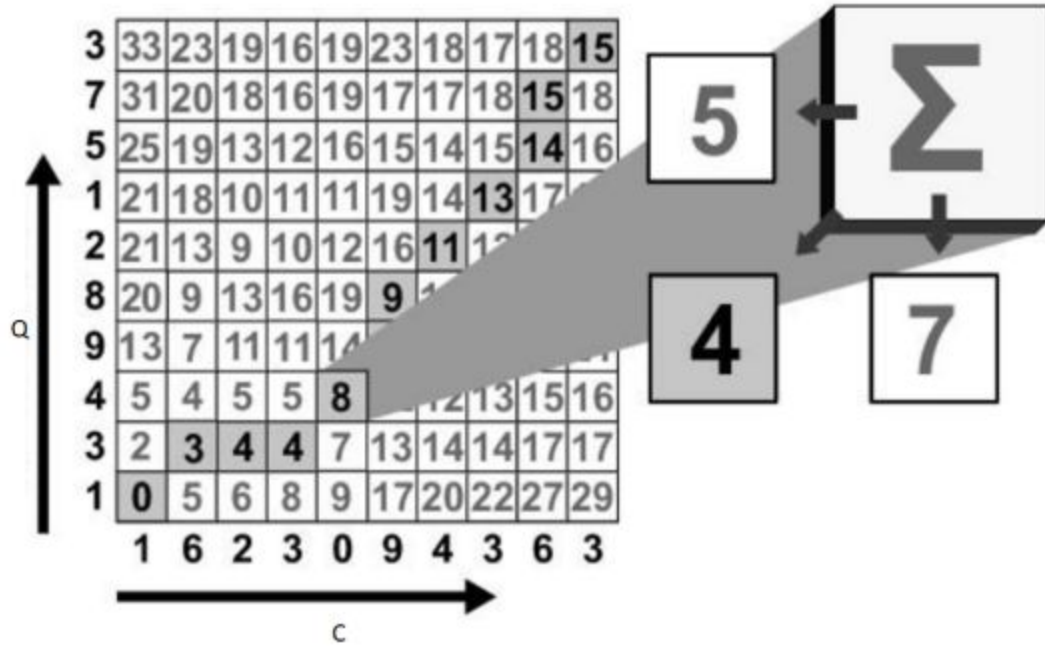
DTW



DTW



DTW



DTW

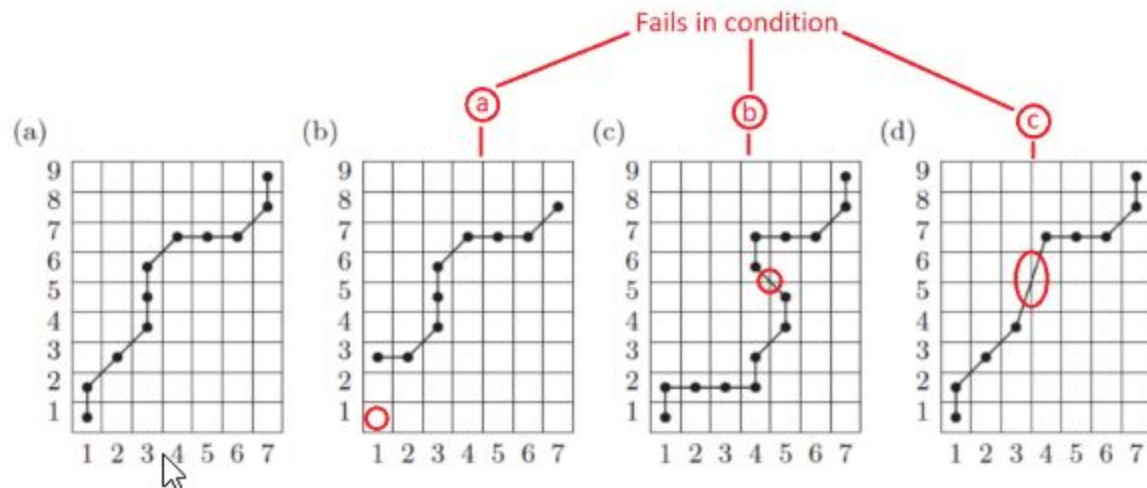


Illustration of paths of index pairs for some sequence. (a) Admissible warping path satisfying the boundary, monotonicity, step size conditions. (b) Boundary condition is violated. (c) Monotonicity condition is violated. (d) Step size condition is violated.

Source: Adapted from Müller (2007).



—

WCPMS

Phenology

Land Surface Phenology (LSP) is the term used for vegetation phenology derived from Earth observation (EO) satellite images (Caparros-Santiago, 2021).

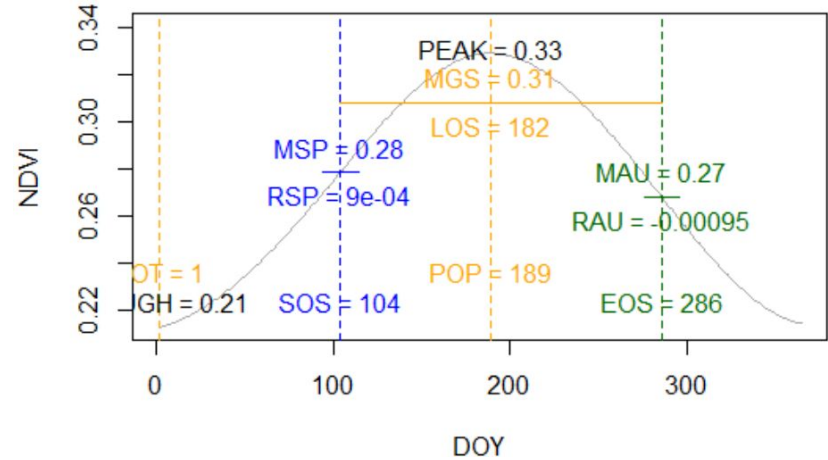
LSP usually uses vegetation indices calculated from spectral properties of satellite images for vegetation change detection (Zeng, 2020).



Phenology metrics

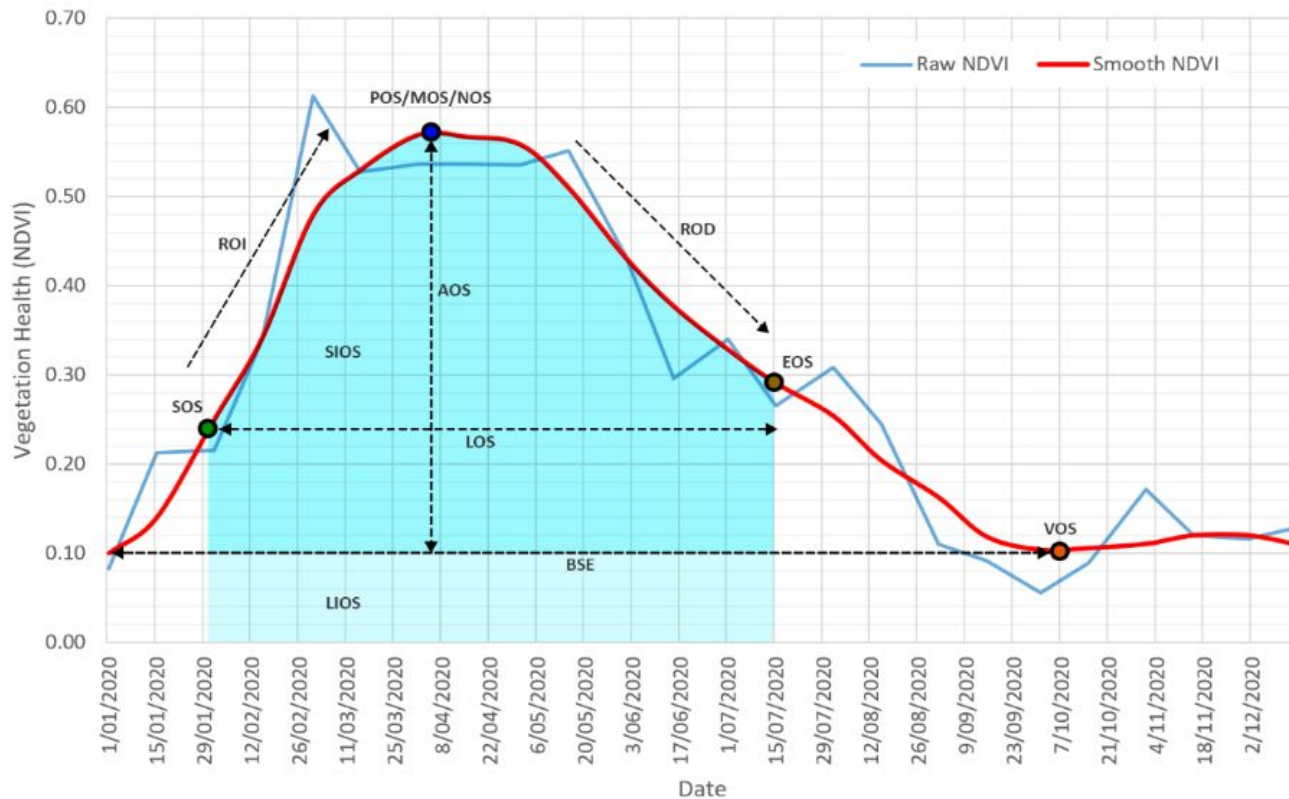
Land Surface Phenology (LSP) is the term used for vegetation phenology derived from Earth observation (EO) satellite images (Caparros-Santiago, 2021).

LSP usually uses vegetation indices calculated from spectral properties of satellite images for vegetation change detection (Zeng, 2020).



Source: (Forkel M, 2015)

Phenology metrics



WCPMS Py Client

A key module of the tool for crop phenology analysis is the Python client.

The wcpms.py library from scratch to facilitate crop phenology extraction operations. This library was developed to be interoperable with other Python libraries, thus enabling users to integrate established libraries into their own workflows for pre- or post-processing and analysis.

```
from wcpms import *

# Defines URL of a instance of the WCPMS
wcpms_url = "http://data.inpe.br/bdc/wcpms"

# Defines a data cube query from Brazil Data Cube (BDC)
datacube = cube_query(
    collection="S2-16D-2",
    start_date="2022-01-01",
    end_date="2022-12-31",
    freq="16D",
    band="NDVI"
)

# Retrieving the phenological metrics
pm = get_phenometrics(
    url = wcpms_url,
    cube = datacube,
    latitude=-13.67705998895876,
    longitude=-52.47134616677471
)

# Visualizing the phenological metrics
print(pm)
```



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RESEARCH ARTICLE



A tool for crop phenology metrics analysis from big Earth observation data

Gabriel Sansigolo^a, Karine Reis Ferreira^a, Gilberto Ribeiro De Queiroz^a,
Thales Körting^a, Leonardo Pereira Garcia Leão^b and Marcos Adami^b

^aEarth Observation and Geoinformatics Division, National Institute for Space Research (INPE), Sao Jose dos Campos, Brazil; ^bSerzedello Corrêa Institute, Tribunal de Contas da União (TCU), Brasília, Brazil

ABSTRACT

Phenological metrics are a set of measurements obtained from Earth observation (EO) satellite image time series that allow the estimation of phenological stages. These include indicators like the start of the greening season, the onset of senescence, and the growing season length. They are useful for crop monitoring. Today, large volumes of images are produced and made available by different EO satellites. These large EO data sets pose a challenge for storage and processing systems, exceeding the capacity of personal computers to handle them. This paper presents a free and open-source tool for phenological metrics analysis from large EO image collections that runs on server-side infrastructure and does not require local data downloads. The Web Crop Phenology Metrics Service (WCPMS) is the core of this tool, designed to estimate phenological metrics as a web service. The tool extracts phenological metrics associated with spatial locations, based on the Brazil Data Cube (BDC) platform. It calculates phenological metrics from data cubes of distinct remote sensing image collections. The potential of the tool is shown through an experiment estimating soybean sowing dates using phenological metrics compared with field data obtained in the Central-South region of Brazil.

ARTICLE HISTORY

Received 6 October 2025
Accepted 7 February 2026

KEYWORDS

Big data; phenological metrics; Python package; remote sensing; spatio-temporal analysis; satellite image time series; web service



Time Series Prediction

Important Concepts: Stationary

Definition

A stationary time series has statistical properties that remain approximately constant over time.

Characteristics

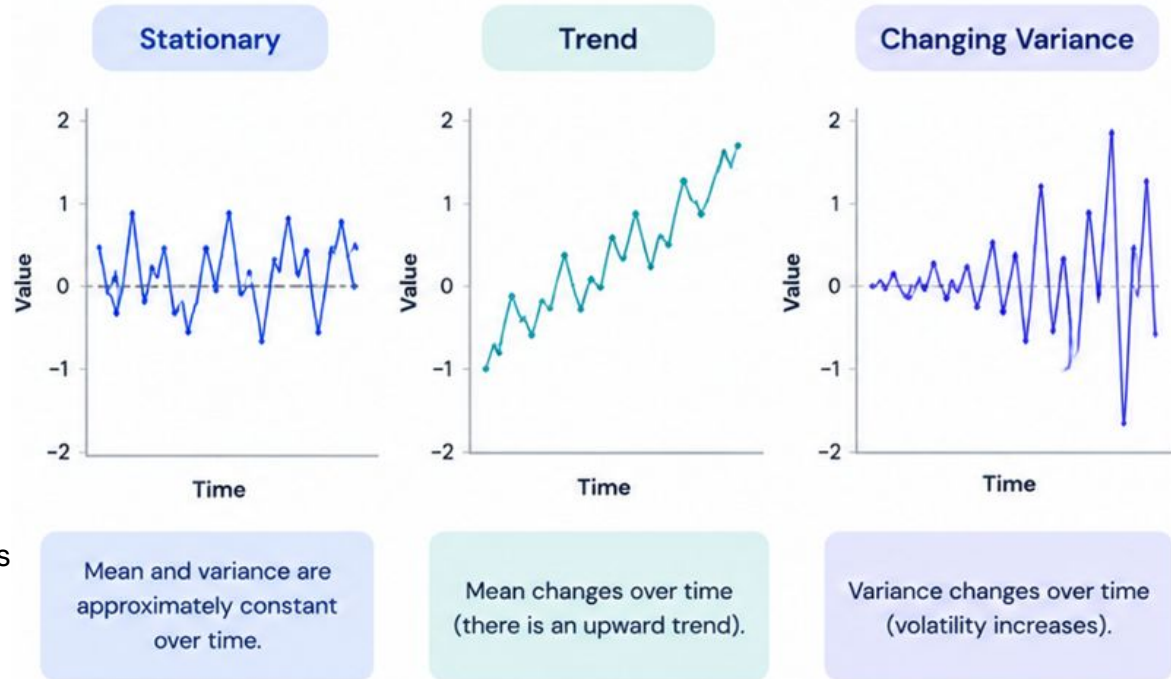
- Constant mean
- Constant variance
- Constant autocorrelation structure

Importance of stationarity

Many classical forecasting methods (e.g., ARIMA) assume stationarity.

When this assumption is violated, predictions may become unreliable.

Examples of Time Series



Only the first series is approximately stationary.

Important Concepts: Autocorrelation

Definition

Autocorrelation measures how similar a time series is to itself at different time lags.

Exemple

If today's river discharge is high, tomorrow's discharge is also likely to be high.

Importance of autocorrelation

Temporal dependence makes forecasting possible.

Without autocorrelation, past observations provide little information about future values.

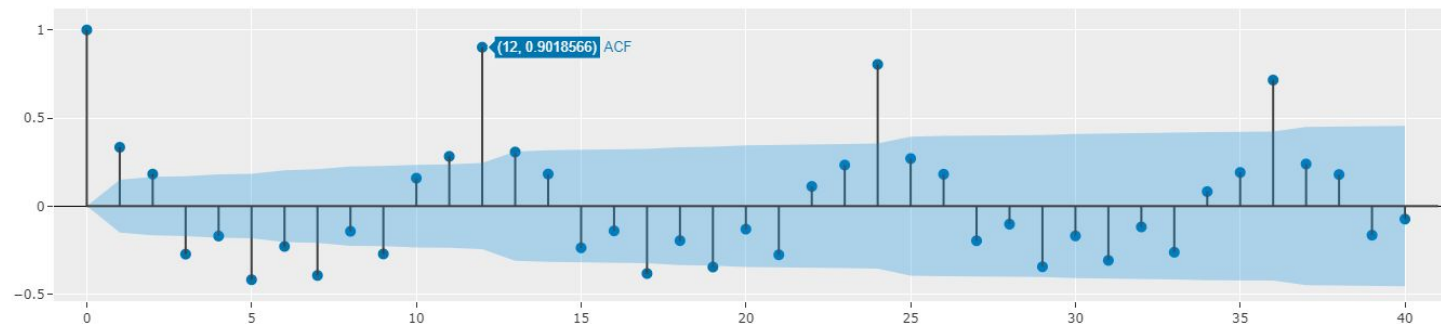
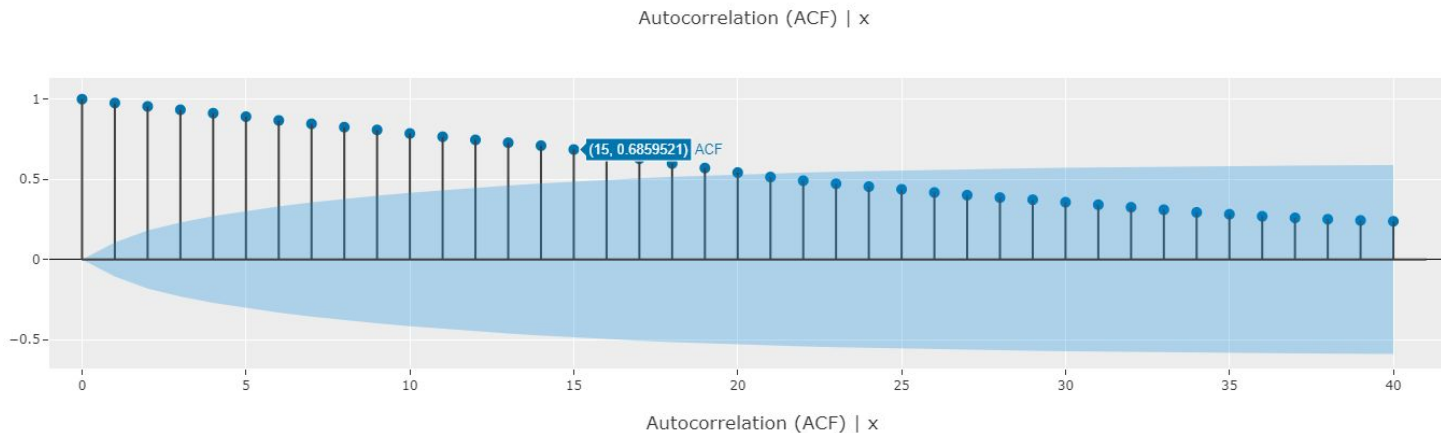
Autocorrelation explains why lag variables often become the most important predictors in machine learning models.

Correlation usually decreases as the time lag increases.

Important Concepts: Autocorrelation

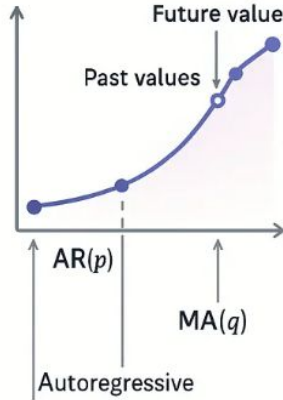
How to interpret the ACF plot

- The blue shaded area represents the **95% confidence interval**.
- **Bars inside** the shaded area are **not statistically significant**.
- **Bars outside** the shaded area indicate **statistically significant autocorrelation** at that lag.



Classical statistics models

Autoregressive Integrated Moving Average ARIMA



$AR(p)$
AutoRegressive

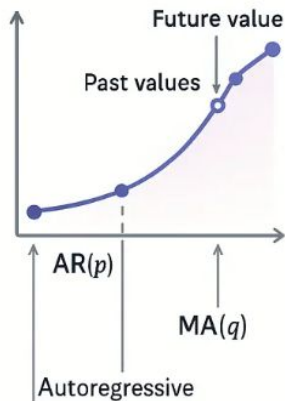
$I(d)$
Integrated

$MA(q)$
Moving Average

Classical statistics models

Autoregressive Integrated Moving Average

ARIMA



AR(p)
AutoRegressive

I(d)
Integrated

MA(q)
Moving Average

```
import numpy as np
from statsmodels.tsa.arima.model import ARIMA
import matplotlib.pyplot as plt
```

```
# 1. Simulate AR(1) time series
```

```
np.random.seed(42)
n = 100
e = np.random.normal(0, 1, n)
y = np.zeros(n)
for t in range(1, n):
    y[t] = 0.5 * y[t-1] + e[t]
```

```
# 2. Fit ARIMA(1,0,0)
```

```
model = ARIMA(y, order=(1, 0, 0))
fit = model.fit()
print(fit.summary())
```

```
# 3. Forecast next 10 steps
```

```
forecast_steps = 10
forecast_result =
fit.get_forecast(steps=forecast_steps)
forecast_mean = forecast_result.predicted_mean
forecast_ci = forecast_result.conf_int()
```

```
# 4. Plot
```

```
plt.figure(figsize=(12, 6))
plt.plot(y, label="Original Series")
plt.plot(fit.fittedvalues, label="Fitted Values",
linestyle='--')
```

```
# Forecast line
```

```
forecast_index = np.arange(len(y), len(y) +
forecast_steps)
plt.plot(forecast_index, forecast_mean,
label="Forecast", color='green')
```

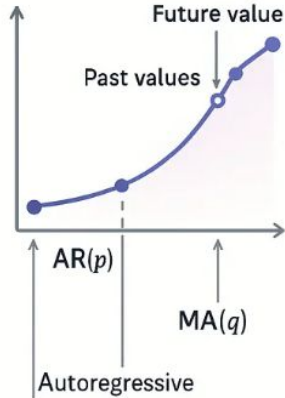
```
# Confidence interval shading
```

```
plt.fill_between(forecast_index,
forecast_ci[:, 0],
forecast_ci[:, 1],
color='green', alpha=0.2, label="95%
Confidence Interval")
```

```
plt.title("ARIMA(1,0,0): Fit and 10-Step Forecast")
plt.xlabel("Time")
plt.ylabel("Value")
plt.legend()
plt.tight_layout()
plt.show()
```

Classical statistics models

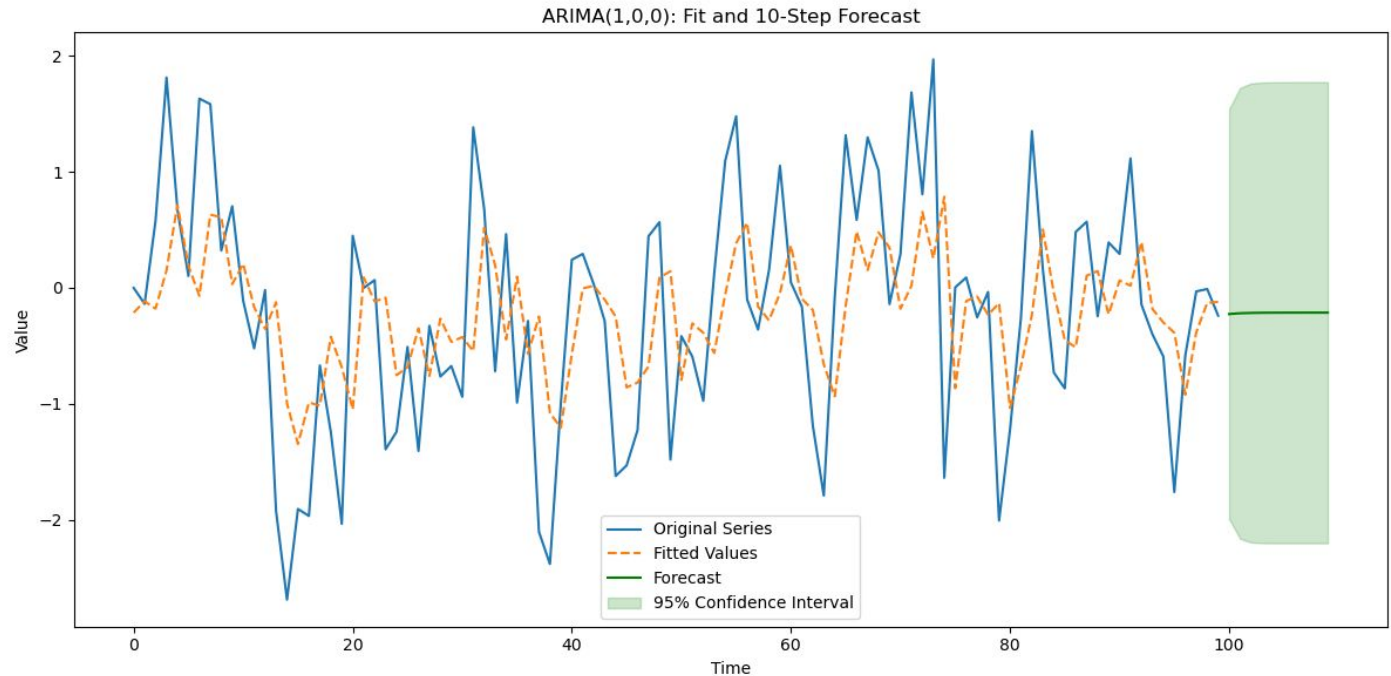
Autoregressive Integrated Moving Average ARIMA



$AR(p)$
AutoRegressive

$I(d)$
Integrated

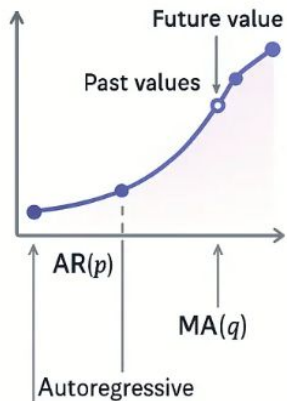
$MA(q)$
Moving Average



Classical statistics models

Autoregressive Integrated Moving Average

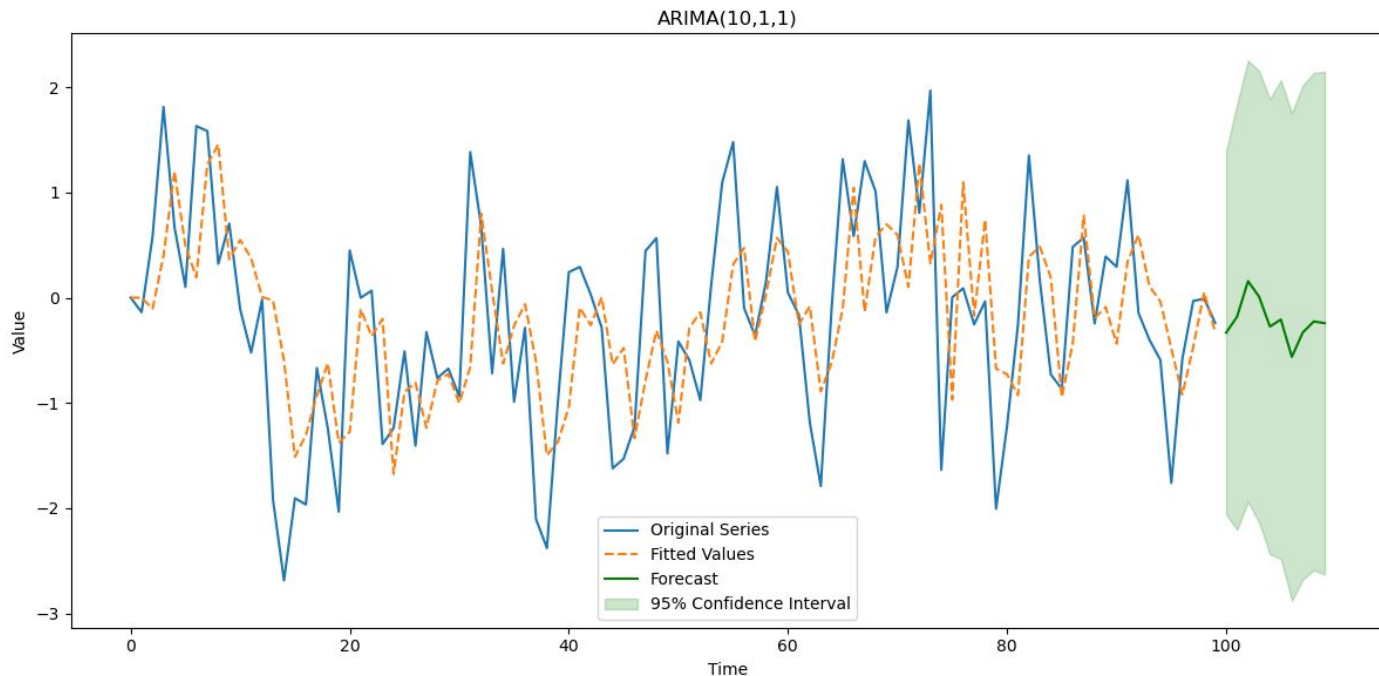
ARIMA



$AR(p)$
AutoRegressive

$I(d)$
Integrated

$MA(q)$
Moving Average



Classical statistics models

Seasonal ARIMA

SARIMA

SARIMA (p, d, q) (P, D, Q, s) where:

- **p, d, q** = non-seasonal ARIMA terms
- **P, D, Q** = seasonal terms
- **s** = length of seasonality (12 for monthly data)



Classical statistics models

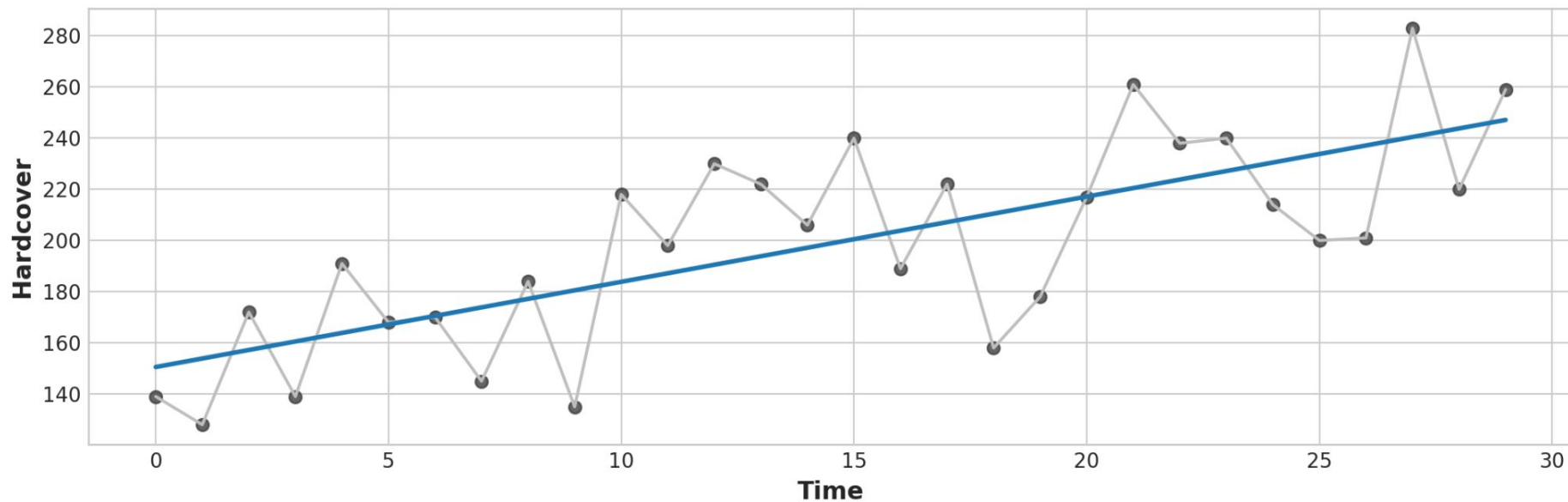
AutoARIMA

Augmented Dickey–Fuller (ADF)
Kwiatkowski–Phillips–Schmidt–Shin (KPSS)
Phillips–Perron (PP)

Akaike Information Criterion (AIC)
Corrected Akaike Information Criterion (AICc)
Bayesian Information Criterion (BIC)

Time Series Prediction Linear Regression

Time Plot of Hardcover Sales



Time Series Prediction

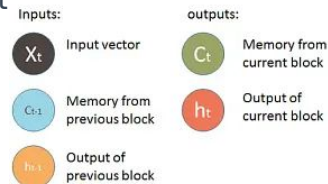
Long Short-Term Memory

LSTM

Recurrent Neural Network (RNN)



X_t is the input of the current time step
 h_{t-1} is the output from the previous LSTM unit
 C_{t-1} is the “memory” of the previous unit



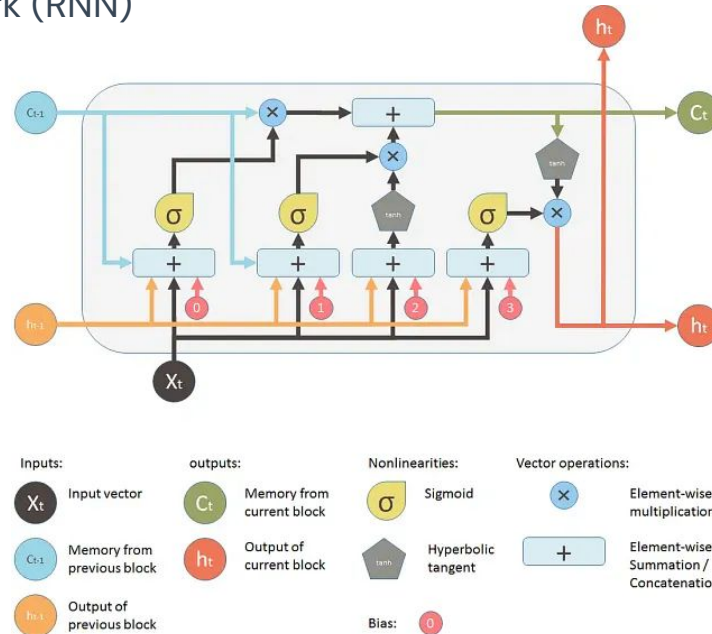
h_t is the output of the current network
 C_t is the memory of the current unit

Time Series Prediction

Long Short-Term Memory

LSTM

Recurrent Neural Network (RNN)

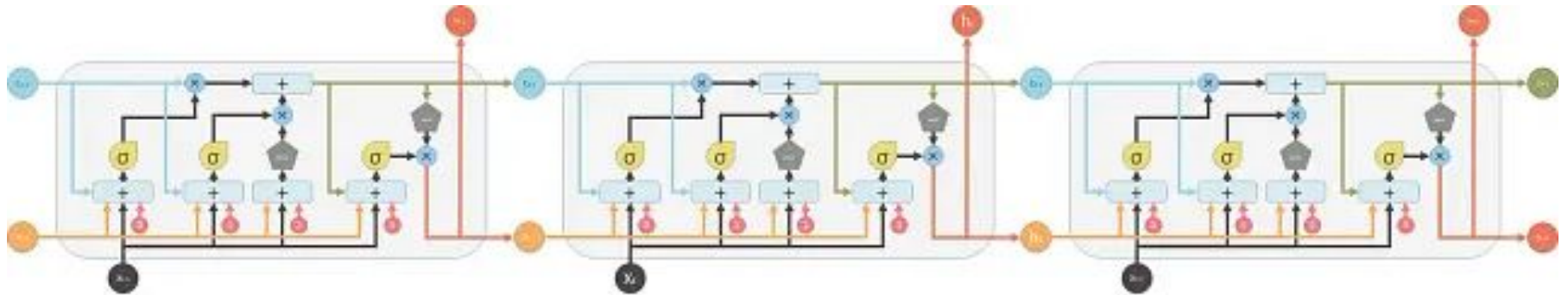


Time Series Prediction

Long Short-Term Memory

LSTM

Recurrent Neural Network (RNN)





Temporal validation

Temporal validation

In time series, observations **must remain in chronological order**.

Random splitting can allow **future information** to influence model training.

 **Wrong**

Train and test observations are randomly mixed.



Future information leaks into training.

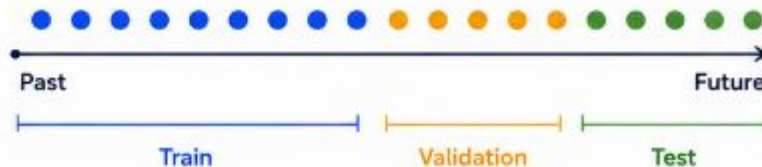
Time (chronological order) →



 **Correct**

Respect the chronological order.

Time (chronological order) →



Always evaluate the model on observations that occur after the training period.

Walk-Forward Validation

Definition

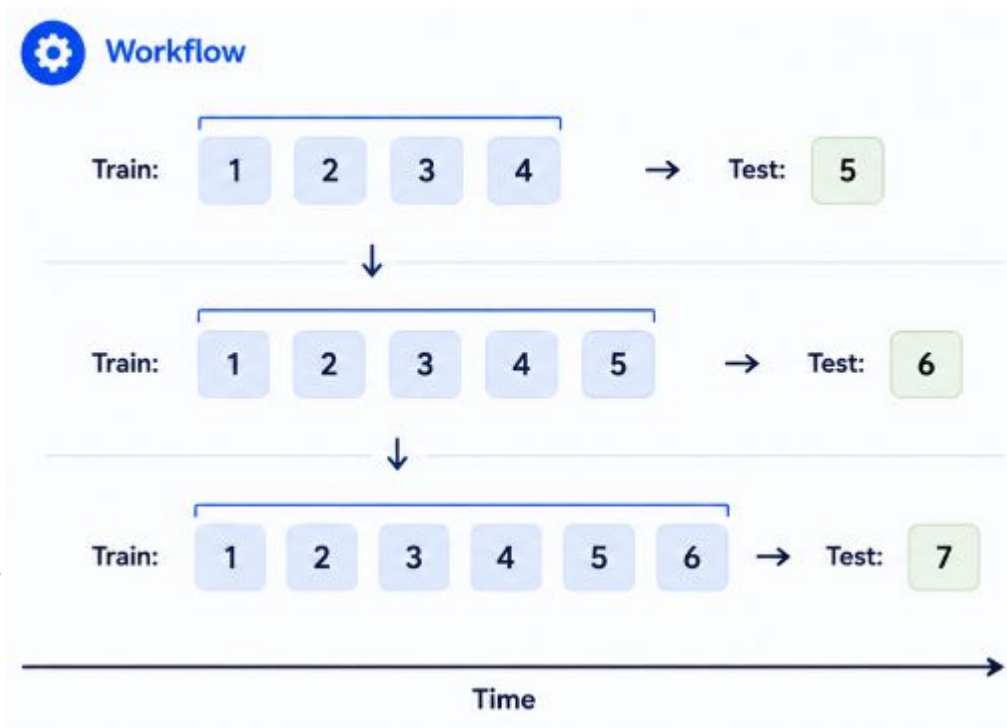
A validation strategy in which the model is repeatedly trained on past observations and evaluated on the next unseen time period.

Advantages

- Preserves chronological order
- Simulates real-world forecasting
- Evaluates model stability over time
- Reduces the risk of data leakage

Limitation

The model must be trained several times, increasing computational cost.



TimeSeriesSplit



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[ShuffleSplit](#)

[StratifiedGroupKFold](#)

[StratifiedKFold](#)

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[TimeSeriesSplit](#)

[check_cv](#)

[train_test_split](#)

[GridSearchCV](#)

[HalvingGridSearchCV](#)

[HalvingRandomSearchCV](#)

[ParameterGrid](#)

[ParameterSampler](#)

[RandomizedSearchCV](#)

[FixedThresholdClassifier](#)

[TunedThresholdClassifierCV](#)

[cross_val_predict](#)

[cross_val_score](#)

[Home](#) > [API Reference](#) > [sklearn.model_selection](#) > **TimeSeriesSplit**

TimeSeriesSplit

```
class sklearn.model_selection.TimeSeriesSplit(n_splits=5, *,  
max_train_size=None, test_size=None, gap=0)
```

[\[source\]](#)

Time Series cross-validator.

Provides train/test indices to split time-ordered data, where other cross-validation methods are inappropriate, as they would lead to training on future data and evaluating on past data. To ensure comparable metrics across folds, samples must be equally spaced. Once this condition is met, each test set covers the same time duration, while the train set size accumulates data from previous splits.

This cross-validation object is a variation of [KFold](#). In the k -th split, it returns the first k folds as the train set and the $(k+1)$ -th fold as the test set.

Note that, unlike standard cross-validation methods, successive training sets are supersets of those that come before them.



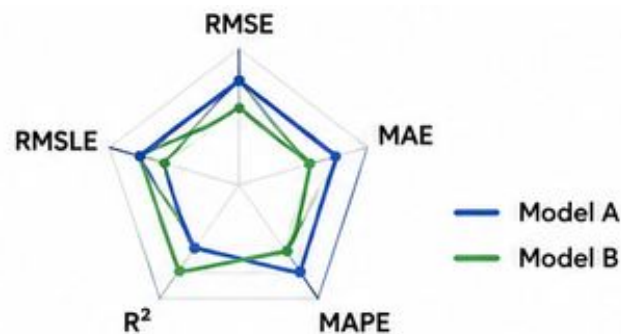
Metrics

Traditional Regression Metrics

Regression performance metrics are based on the difference between the **observed** and **predicted** values:

$$e = y - \hat{y}$$

- MSE (Mean Squared Error)
- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- MAPE (Mean Absolute Percentage Error)
- RMSLE (Root Mean Squared Logarithmic Error)
- R^2 (Coefficient of Determination)



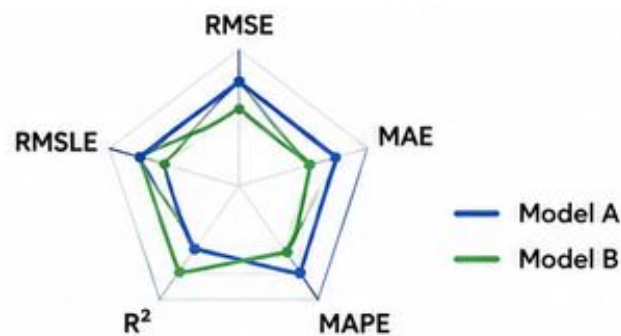
Multiple metrics reveal different strengths and weaknesses of the model.

Time Series Forecast Metrics

Regression performance metrics are based on the difference between the **observed** and **predicted** values:

$$e = y - \hat{y}$$

- **MSE (Mean Squared Error)**
- **RMSE (Root Mean Squared Error)**
- **MAE (Mean Absolute Error)**
- **MAPE (Mean Absolute Percentage Error)**
- **RMSLE (Root Mean Squared Logarithmic Error)**
- **R² (Coefficient of Determination)**



Multiple metrics reveal different strengths and weaknesses of the model.

Time Series Forecast Metrics

sMAPE (Symmetric Mean Absolute Percentage Error)

Measures the percentage error while reducing the sensitivity to very small observed values:

$$\text{sMAPE} = \frac{100}{n} \sum_{i=1}^n \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|) / 2}$$

- A more robust alternative to MAPE for time series forecasting.
- The **lower** the sMAPE, the **better** the model's performance.
- Unlike MAPE, sMAPE is bounded — its **minimum value is 0%**, and its **maximum value is 200%**.

Time Series Forecast Metrics

MASE (Mean Absolute Scaled Error)

Measures prediction error relative to a simple baseline forecast (naïve forecast):

$$\text{MASE} = \frac{\frac{1}{n} \sum |y_i - \hat{y}_i|}{\frac{1}{n-1} \sum |y_t - y_{t-1}|}$$

y_i : Observed value at time i

$\hat{y}_i$: Predicted value at time i

y_t : Observed value at time t

y_{t-1} : Observed value at the previous time step

- A more robust alternative to MAPE for time series forecasting.
- The **lower** the MASE, the **better** the model's performance.
- $\text{MASE} < 1$ means the model outperforms the naïve forecast; $\text{MASE} > 1$ means it performs worse.

Naïve forecast: baseline model that uses the last observed value as the prediction (or the value from one season ago, for seasonal data).

Time Series Forecast Metrics

WAPE (Weighted Absolute Percentage Error)

Measures the total prediction error relative to the total observed magnitude.

$$\text{WAPE} = 100 \cdot \frac{\sum |y_i - \hat{y}_i|}{\sum |y_i|}$$

y_i : Observed value at time i

$\hat{y}_i$: Predicted value at time i

- A volume-weighted alternative to MAPE, less sensitive to distortions from low-magnitude values.
- The **lower** the WAPE, the **better** the model's performance.

Domain-Specific Metrics

Hydrology

NSE (Nash–Sutcliffe Efficiency)

Measures how well the predicted values reproduce the observed variability relative to the mean of the observations.

$$NSE = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

y_i : Observed value

$\hat{y}_i$: Predicted value

$\bar{y}$: mean of the observed values

- A goodness-of-fit metric that compares model predictions to the mean of observed values, widely used in hydrology and environmental modeling.
- The higher the NSE, the better the model's performance (unlike MAPE, sMAPE, WAPE, and MASE, where lower is better).
- Its maximum value is 1 (perfect fit); it has no lower bound, and $NSE = 0$ means the model performs no better than simply predicting the mean.

Best used for: Evaluating streamflow, water quality, and sediment prediction models.

Domain-Specific Metrics

Hydrology

KGE (Kling–Gupta Efficiency)

Evaluates prediction accuracy by combining correlation, variability, and bias into a single metric.

$$KGE = 1 - \sqrt{(r - 1)^2 + (\alpha - 1)^2 + (\beta - 1)^2}$$

r : Pearson correlation coefficient

α : variability ratio

β : bias ratio

- KGE decomposes model performance into three components (correlation, variability, and bias) providing a more balanced evaluation than NSE, which tends to be overly sensitive to peak values.
- The higher the KGE, the better the model's performance (like NSE, and unlike MAPE, sMAPE, WAPE, and MASE).
- Its maximum value is 1 (perfect fit); it has no lower bound, and $KGE = -0.41^*$ is often used as the benchmark for "no skill" (equivalent to always predicting the mean).

*Knoben, W. J., Freer, J. E., & Woods, R. A. (2019). Inherent benchmark or not? Comparing Nash–Sutcliffe and Kling–Gupta efficiency scores. *Hydrology and Earth System Sciences*, 23(10), 4323–4331.

Domain-Specific Metrics

Hydrology

PBIAS (Percent Bias)

Measures the average tendency of the model to overestimate or underestimate observations.

$$PBIAS = 100 \cdot \frac{\sum(\hat{y}_i - y_i)}{\sum y_i}$$

y_i : Observed value

$\hat{y}_i$: Predicted value

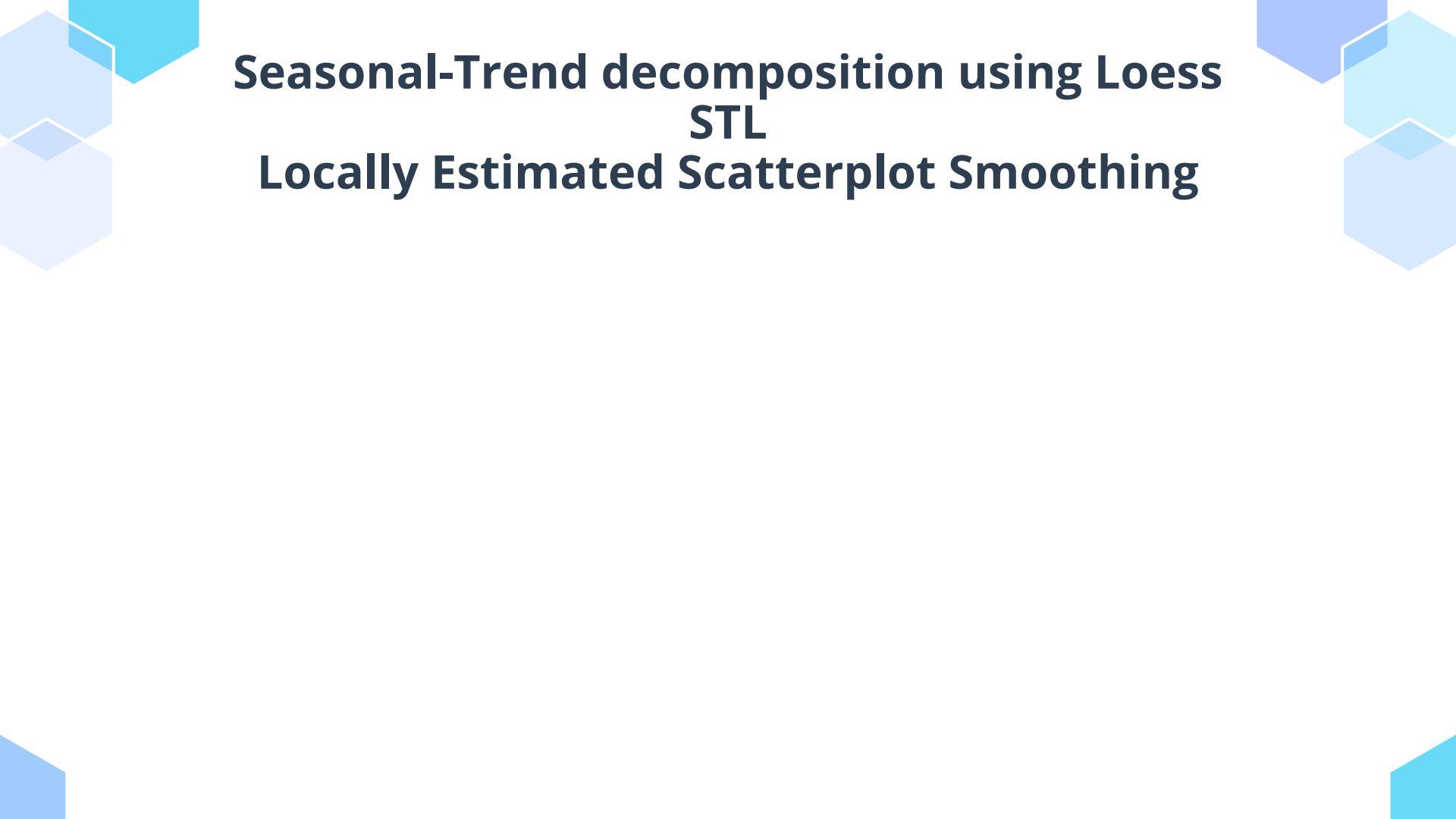
- A bias-focused metric that measures the average tendency of the model to over- or underestimate observed values.
- The closer PBIAS is to 0%, the better the model's performance — it does not measure overall accuracy, only systematic bias.
- Positive PBIAS indicates model underestimation; negative PBIAS indicates model overestimation.

Best used for: Detecting systematic overestimation or underestimation.



—

STL

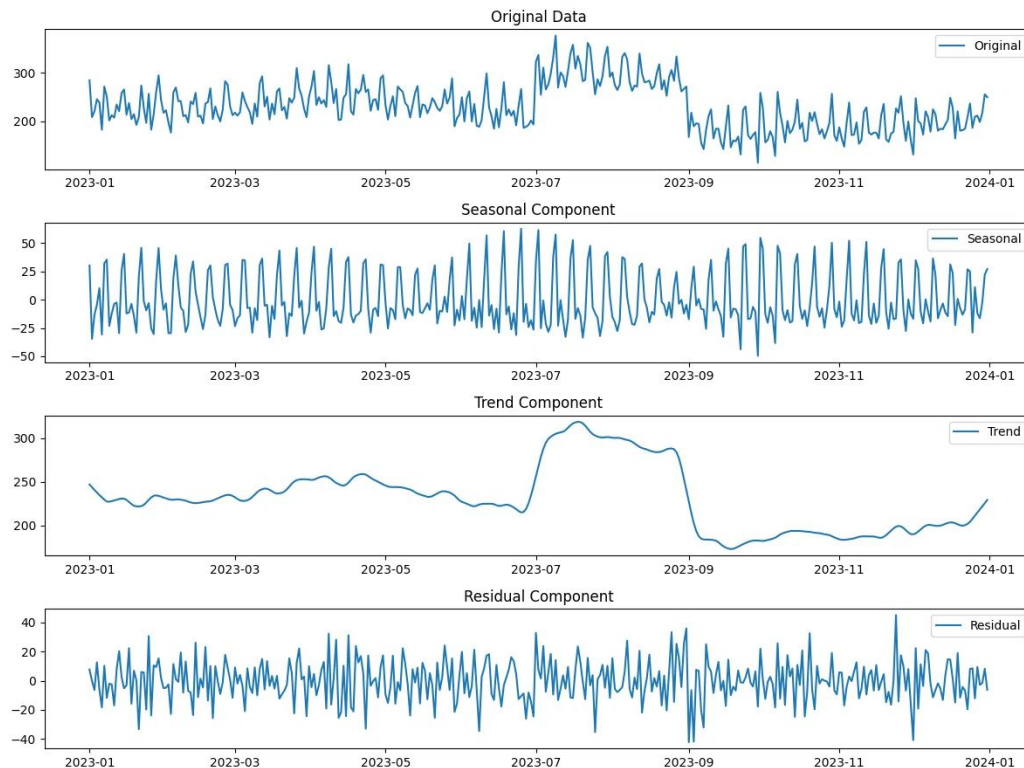


Seasonal-Trend decomposition using Loess

STL

Locally Estimated Scatterplot Smoothing

Seasonal-Trend decomposition using Loess STL Locally Estimated Scatterplot Smoothing





Trend analysis in time series

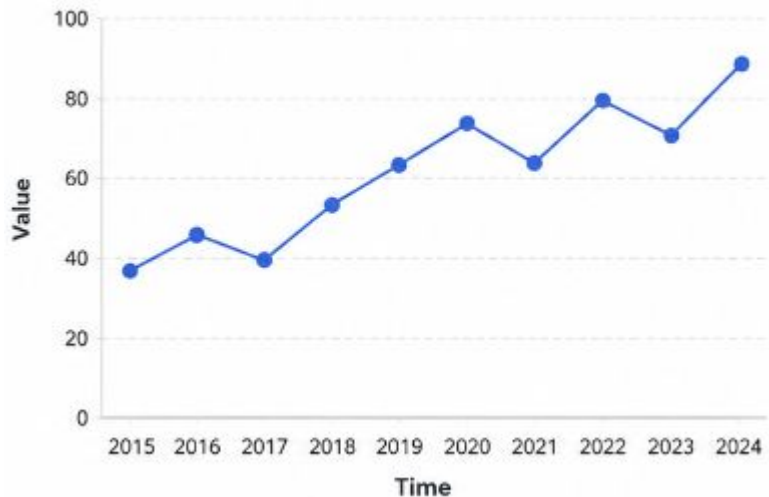
Trend analysis in time series

Many time series are not stationary.

Instead of predicting future values, we may want to answer questions such as:

- 🔍 Is there a **significant trend**?
- ↗ Is the trend **increasing** or **decreasing**?
- 📏 How **strong** is the trend?
- 🕒 Has the behavior of the series **changed** over time?

Example of a Time Series



Different **statistical methods** answer different questions.

Mann-Kendall Trend Test

Definition

A non-parametric statistical test used to determine whether a monotonic trend exists in a time series.

What does it answer?

Is there a statistically significant trend?

Output

- Increasing trend
- Decreasing trend
- No significant trend

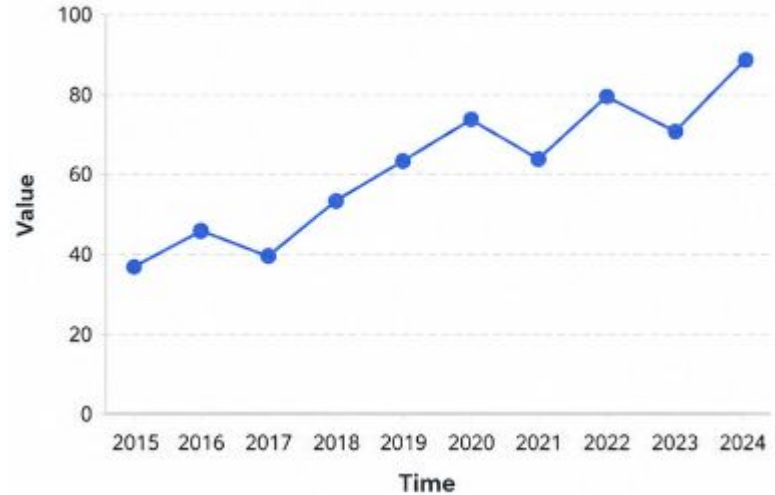
Advantages

- Does not assume normality
- Robust to outliers
- Widely used in hydrology and climatology

Limitation

Analyzes the entire time series as a single period. It does not identify when changes occurred.

Example of a Time Series



Decision

Significant increasing trend
(No significant trend?)

Sen's Slope Estimator

Definition

A non-parametric estimator that measures the magnitude of a monotonic trend.

What does it answer?

How fast is the variable changing?

Example

Variable	Sen's Slope (Trend)
River discharge	+3.2 m ³ /s per year
NDVI	-0.004 year ⁻¹

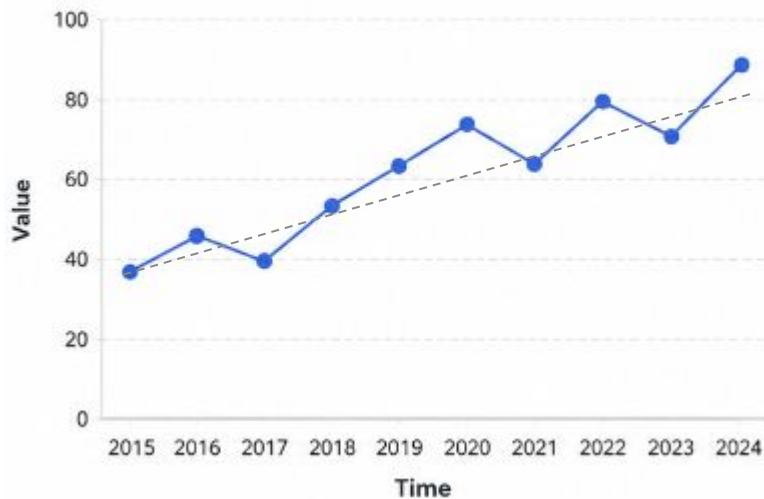
Advantages

- Robust to outliers
- Works well with non-normal data
- Commonly combined with Mann–Kendall

Limitation

Assumes a single trend throughout the entire series.

Example of a Time Series



Sen's Slope = +3.2 m³/s per year
The variable increases on average by **3.2 m³/s** each year.



Mann–Kendall tells **whether** a trend exists. **Sen's Slope** tells **how strong** that trend is.

Pettitt Change Point Test

Definition

A non-parametric test used to identify a single abrupt change point in a time series.

What does it answer?

Did the statistical behavior of the series change?

Typical causes

- Dam construction
- Sensor replacement
- Land use change
- Climate regime shift

Output

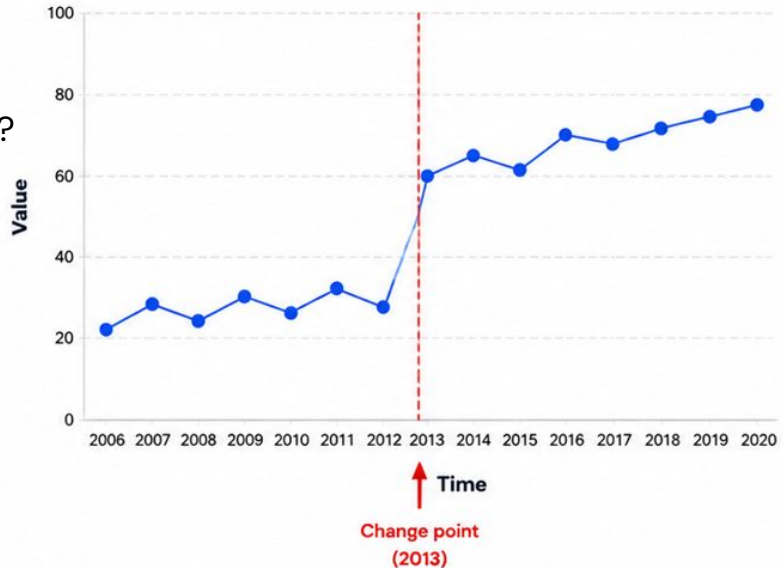
- Estimated change point

Example: 2013

Limitation

Only one change point is detected.

Example Time Series with Change Point





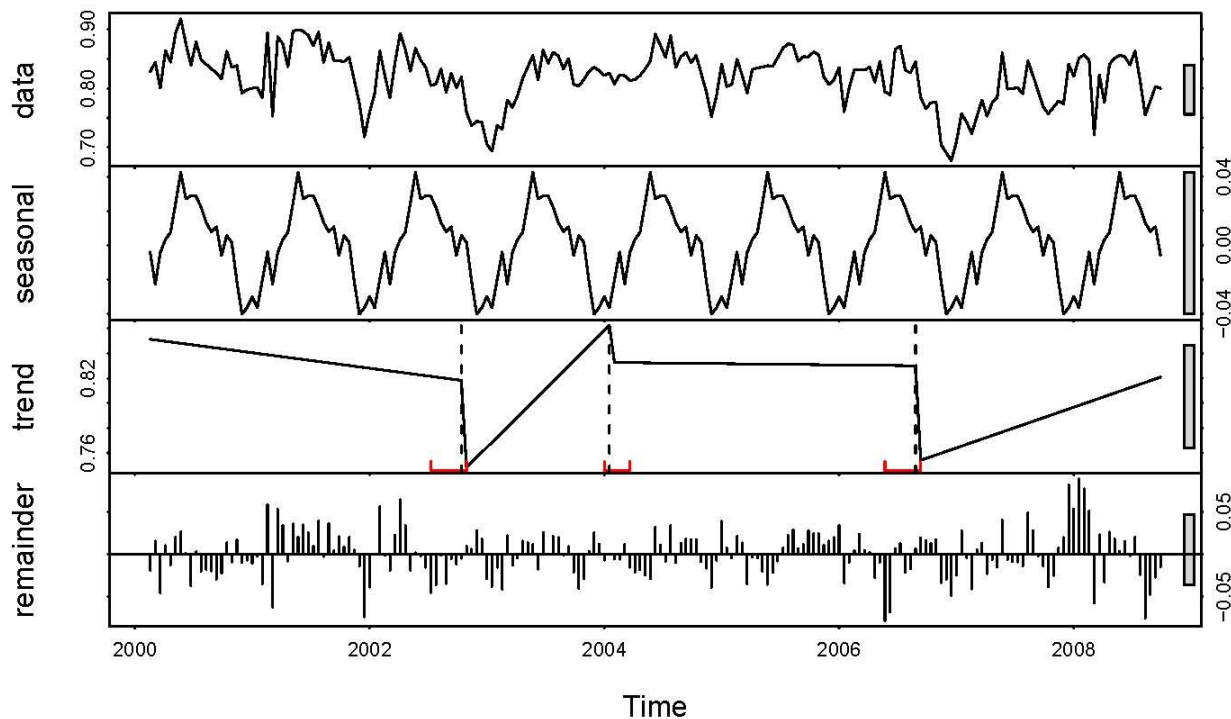
BFAST

Breaks For Additive Season and Trend BFAST

BFAST decomposes the series and detects **multiple breakpoints** and **multiple trend segments**.

Method	Detects Trend	Estimates Trend Magnitude	Detects Change Point	Multiple Changes
Mann-Kendall	✓	✗	✗	✗
Sen's Slope	✗	✓	✗	✗
Pettitt	✗	✗	✓	✗
BFAST	✓	✓	✓	✓

Breaks For Additive Season and Trend BFAST

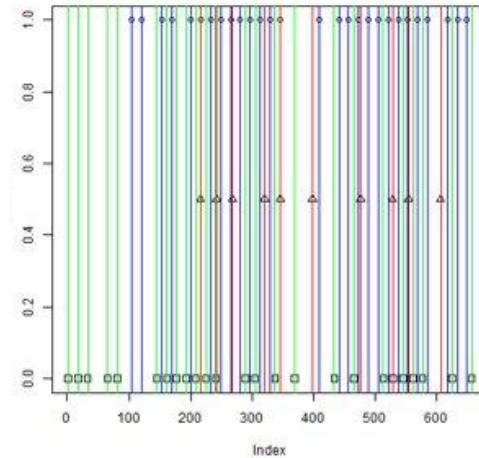
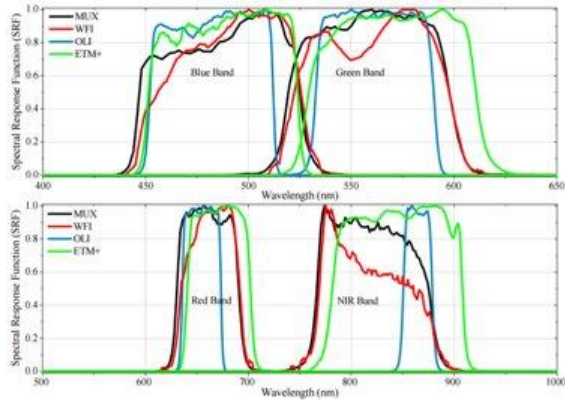
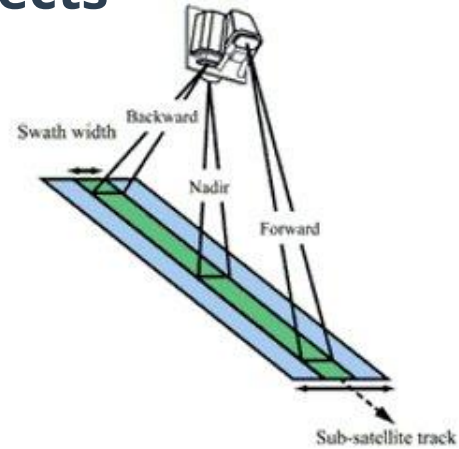
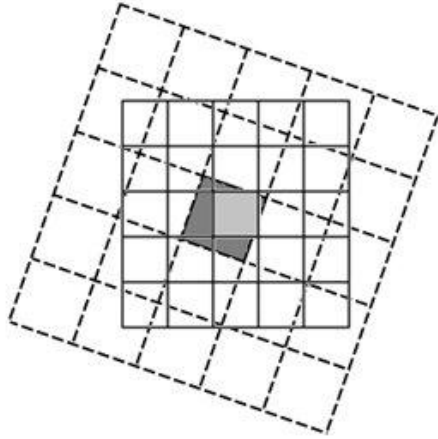




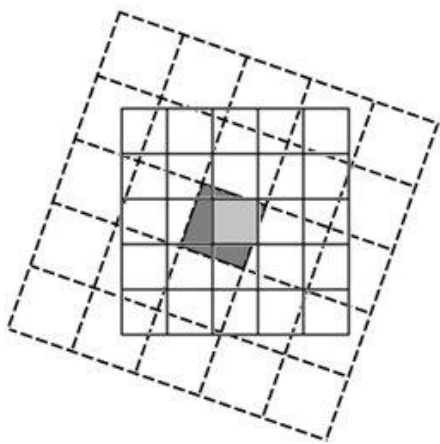
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Thesis

Harmonization Aspects



Spatial Aspect: spatial resolution

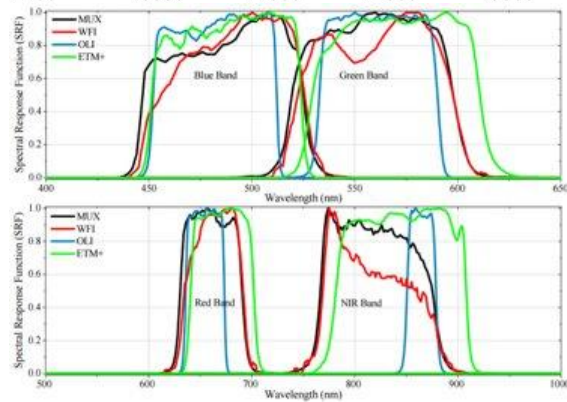
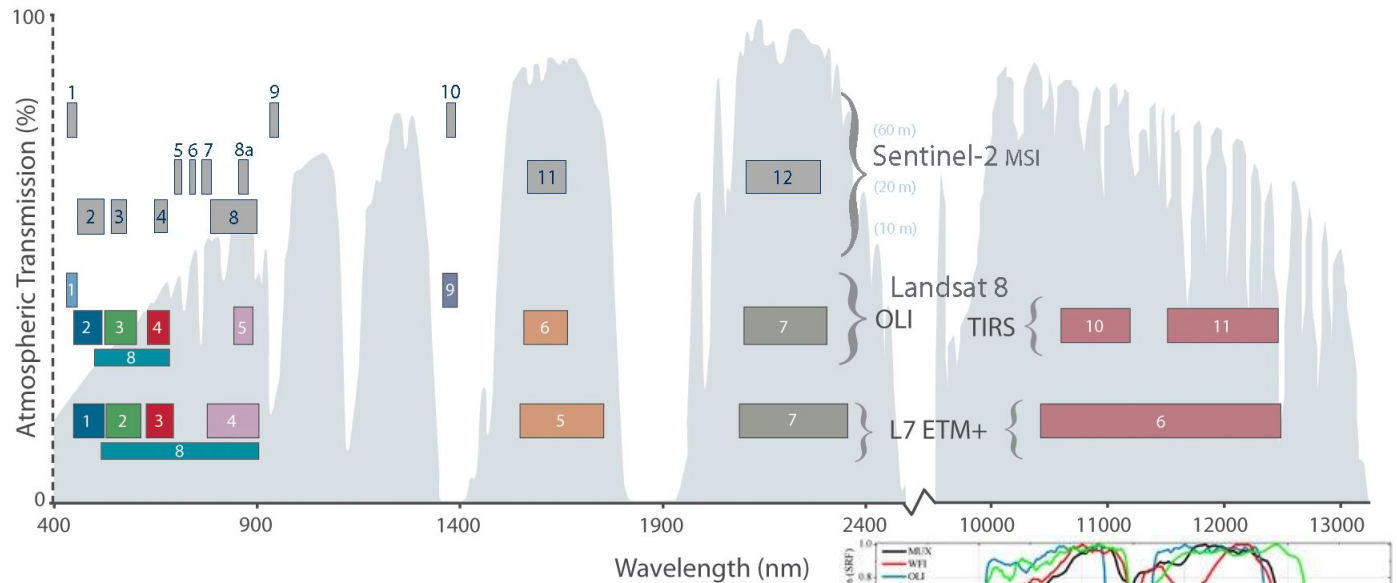


Spatial Aspect: image registration

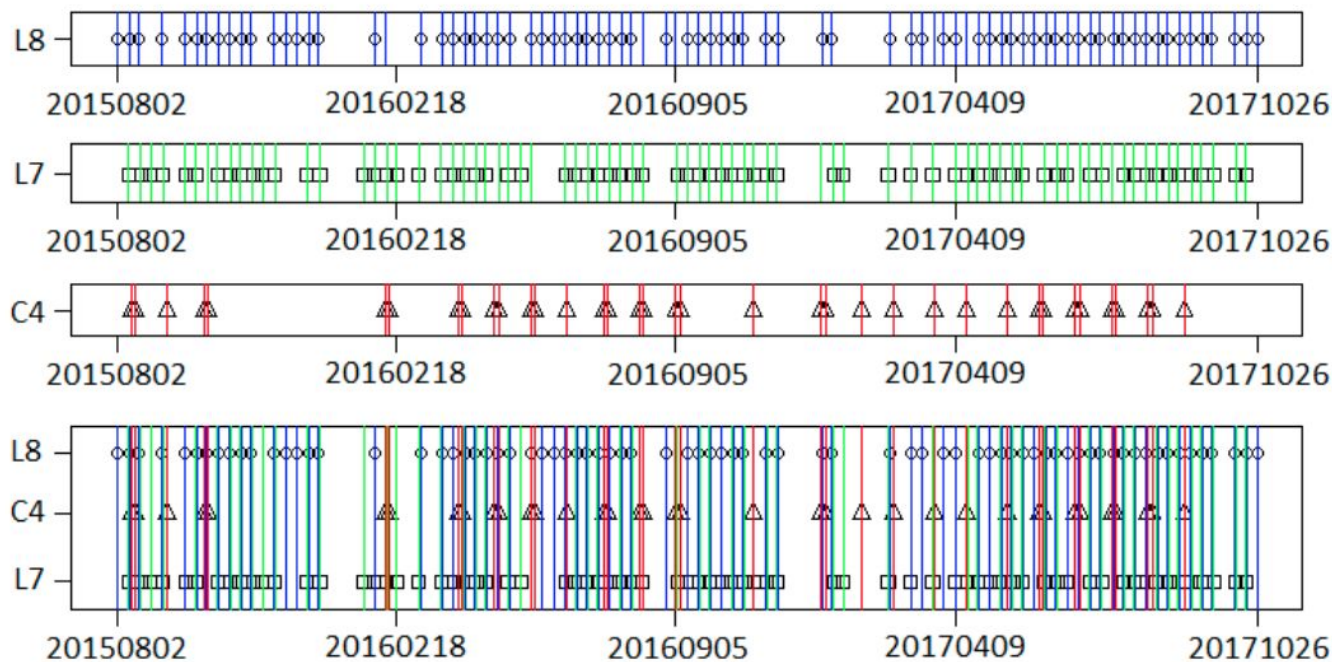
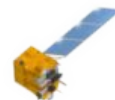
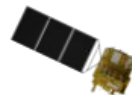


Spectral Aspect

Comparison of Landsat 7 and 8 bands with Sentinel-2



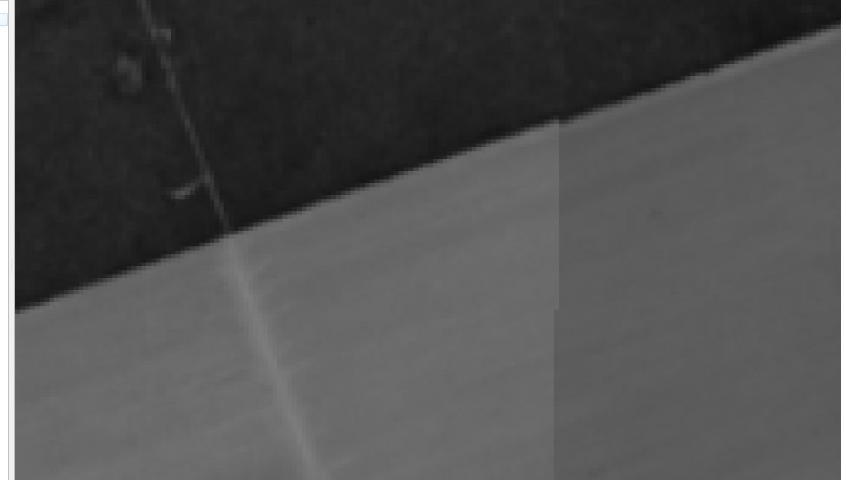
Temporal Aspect



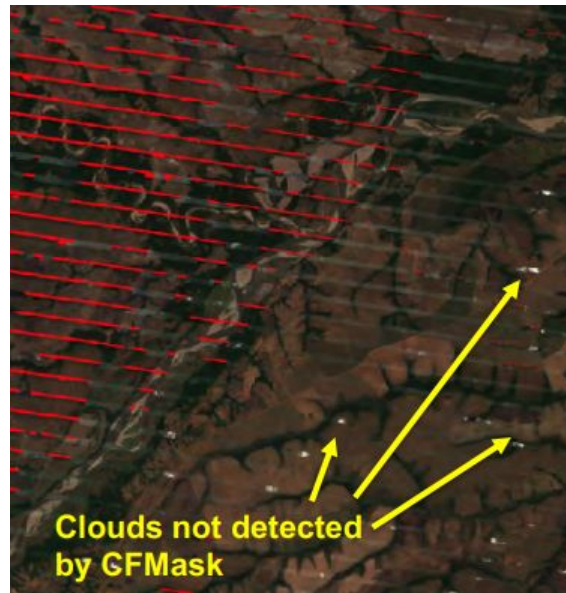
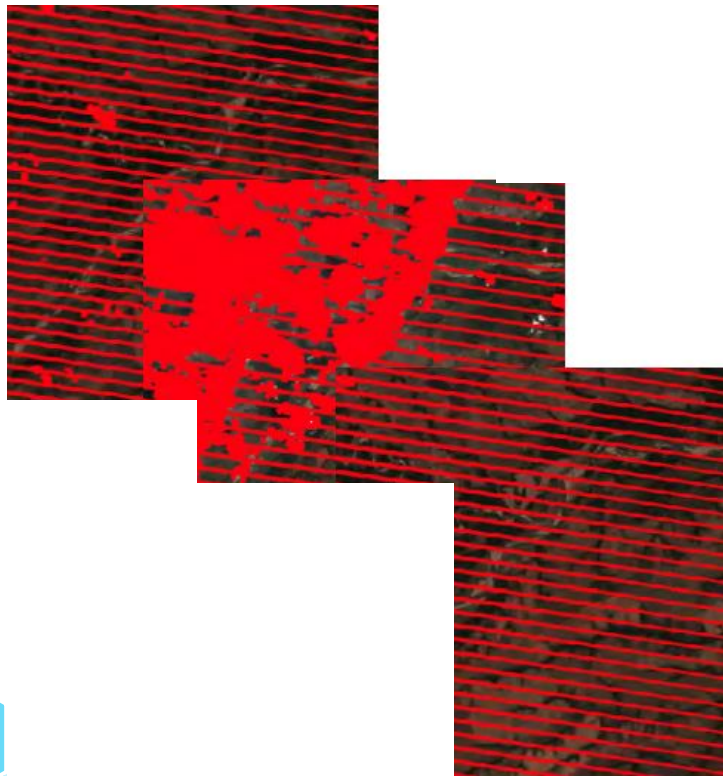
Overlap



- ☐ S10m_089098_2018-09-01_coastal_STACK
- ☐ S10m_089098_2018-09-01_blue_STACK
- ☐ S10m_089098_2018-09-01_green_STACK
- ☒ S10m_089098_2018-09-01_red_STACK



Cloud Mask



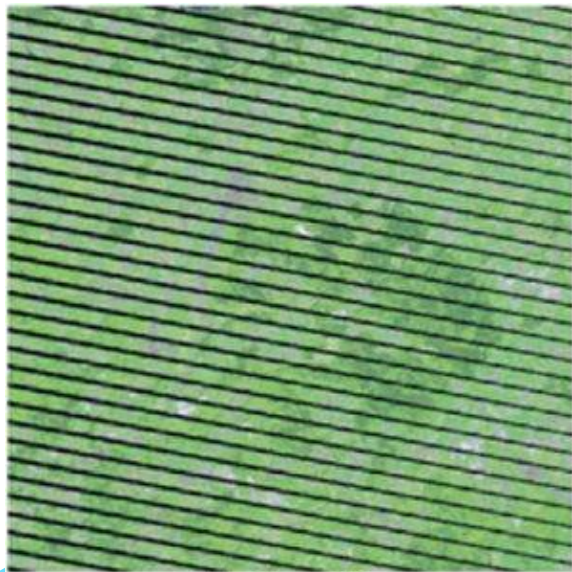
Cloud Shadow



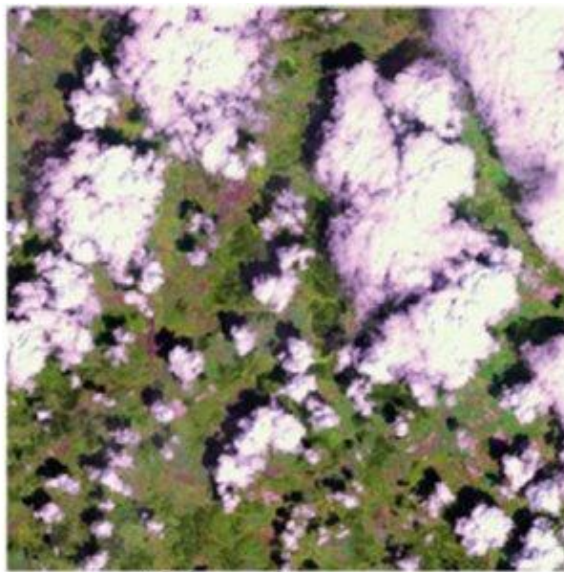
Cloud Shadow



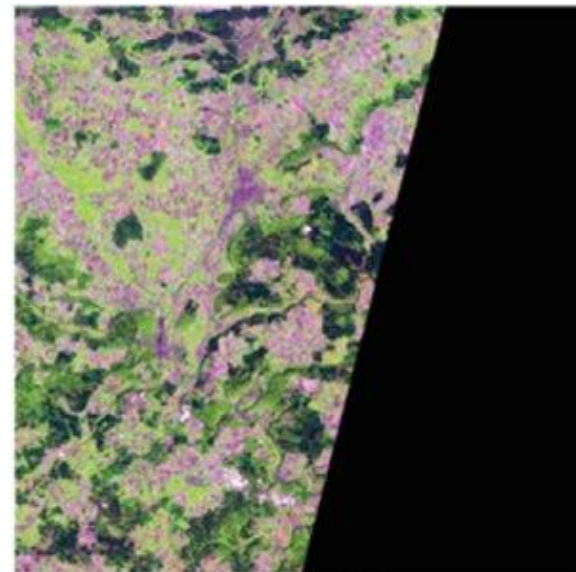
Landsat-7 SLC-off



SLC off

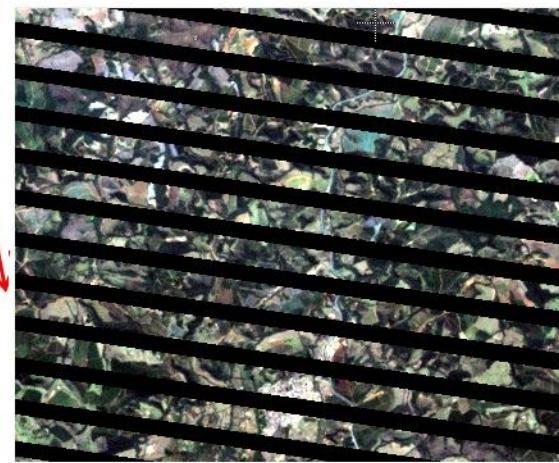
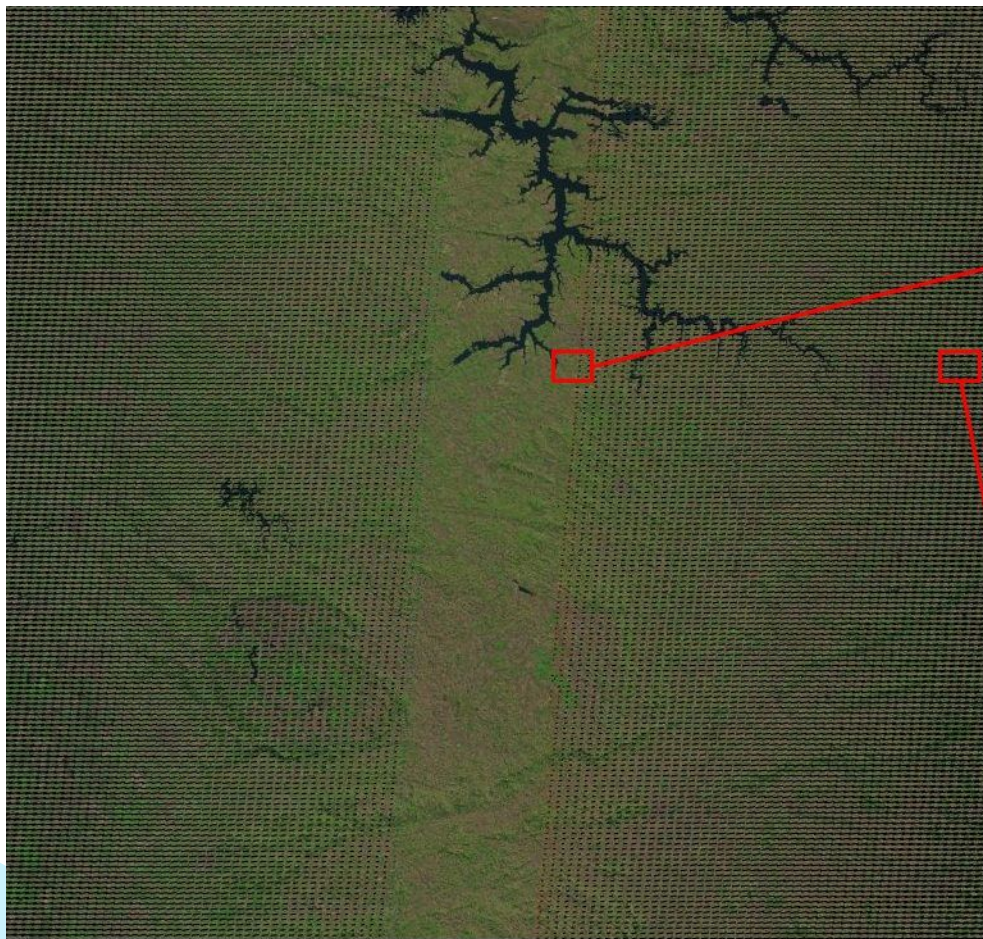


Cloud cover



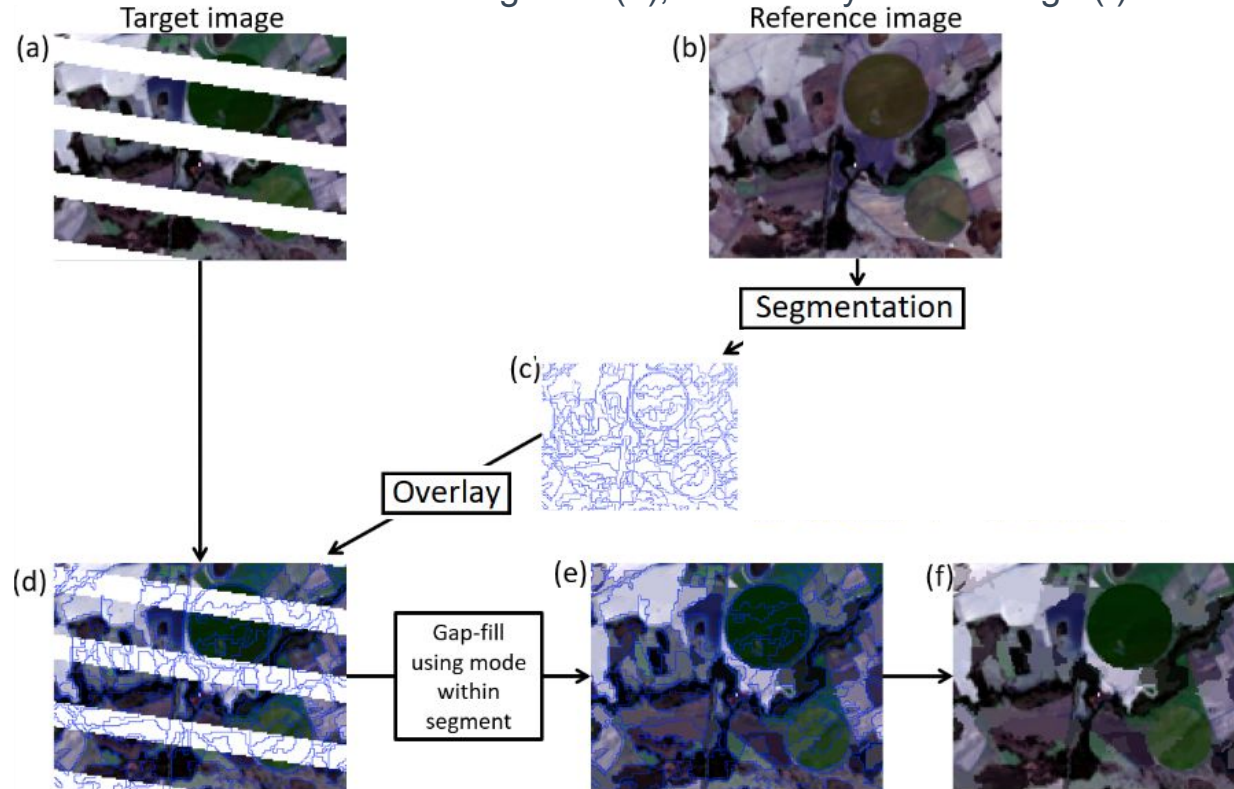
Partial

Landsat-7 SLC-off



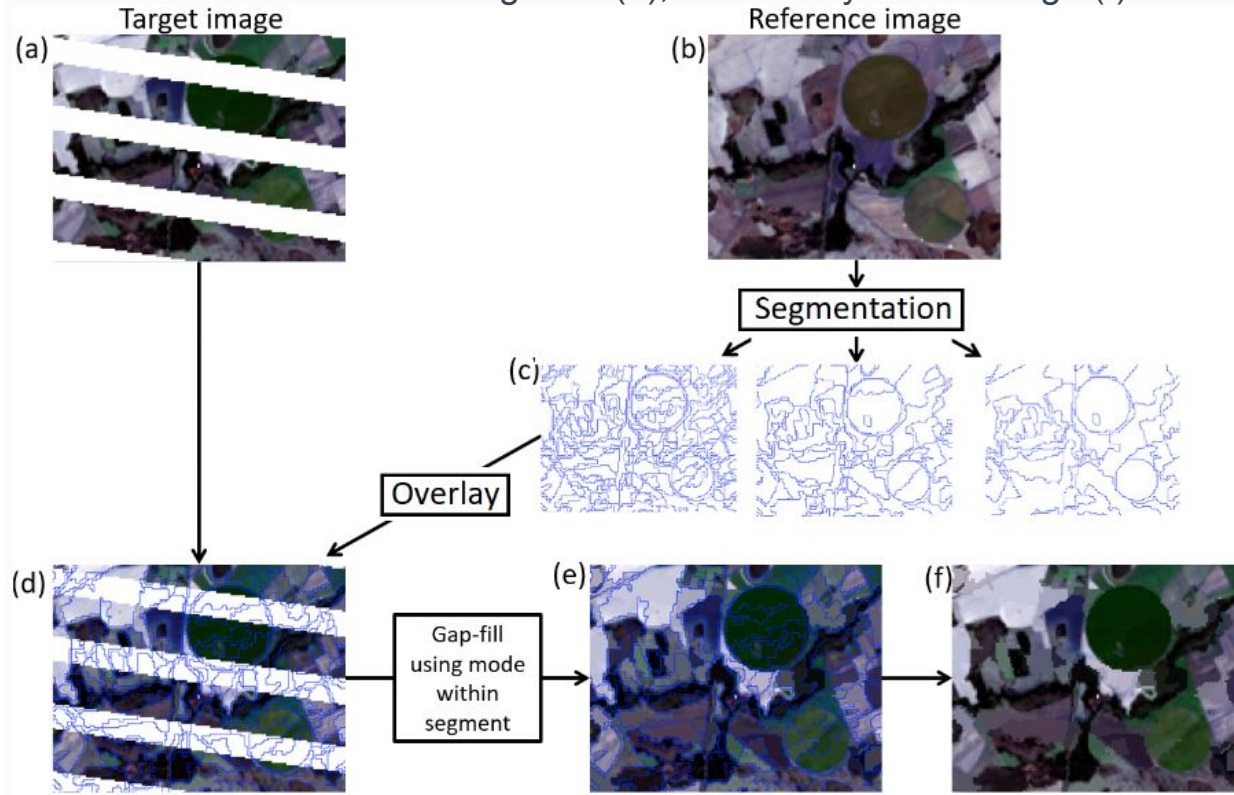
Landsat-7 SLC-off: gap-filling

A target image that contains gaps, in this case a Landsat-7/ETM+ SLC-off (a); a reference image, in this case a Landsat-8/OLI acquired in a close date (b); the reference image is segmented (c); the segmentation is overlaid in the target image (d); the modevalue within the segment is calculated and used to fill missing data (e); resulted synthetic image (f).



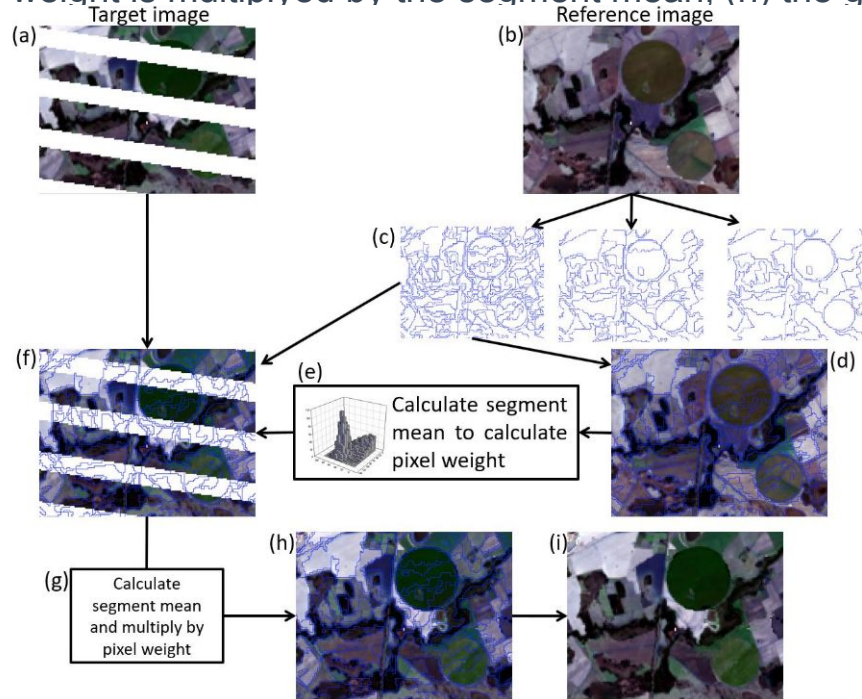
Landsat-7 SLC-off

A target image that contains gaps, in this case a Landsat-7/ETM+ SLC-off (a); a reference image, in this case a Landsat-8/OLI acquired in a close date (b); the reference image is segmented (c); the segmentation is overlaid in the target image (d); the modevalue within the segment is calculated and used to fill missing data (e); resulted synthetic image (f).



Landsat-7 SLC-off

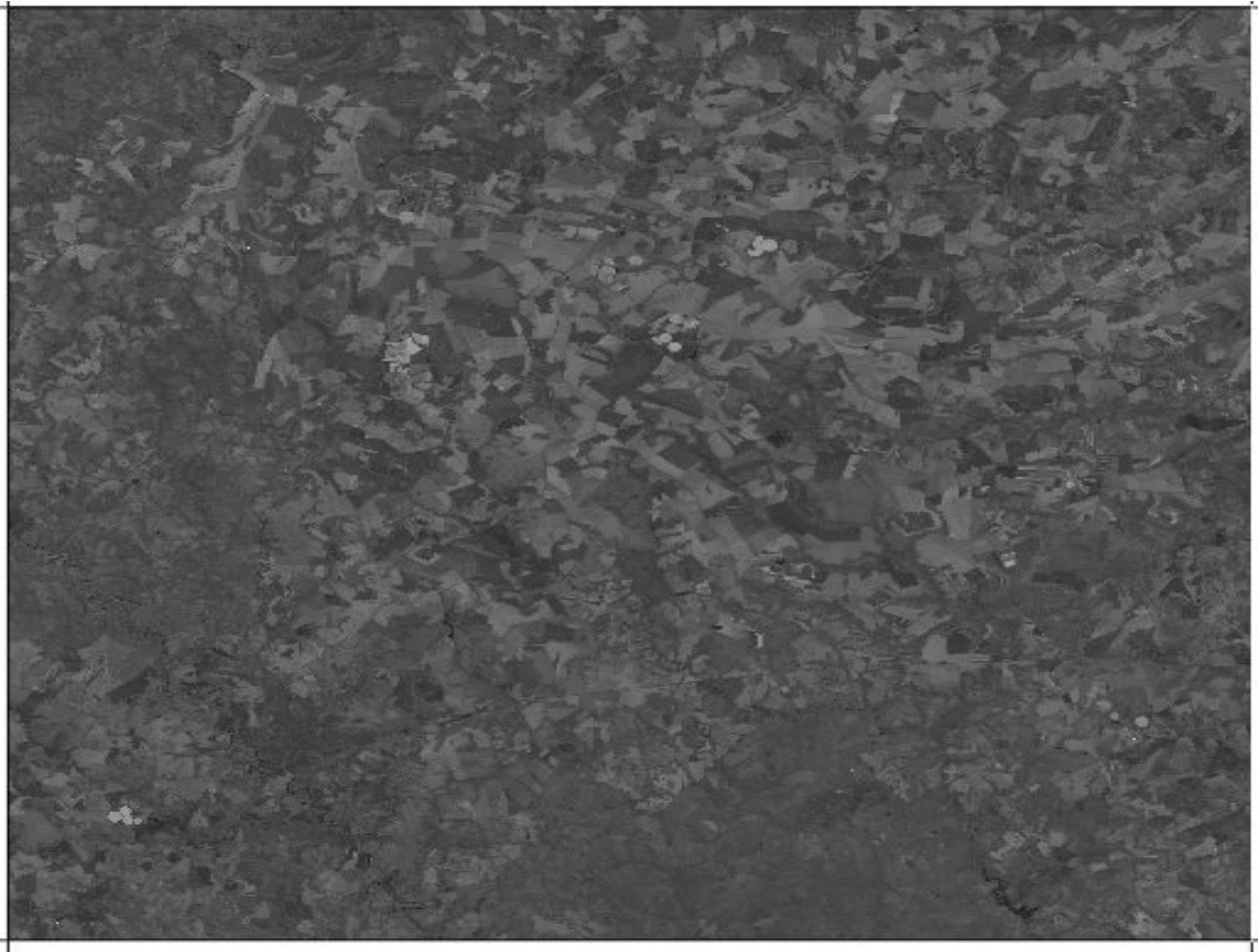
Segmentation approach fills Landsat-7 SLC-off gap images: (a) a target image contains gaps, in this case a Landsat-7 SLC-off; (b) a reference image is used, such as Landsat-5 or Landsat-8 acquired in a close date; (c) the reference image is segmented; (d) the segments are applied in the reference image; (e) the segment mean value is calculated and used to calculate pixel weights; (f) the segmentation is overlaid in the target image and the mean value within the segment is calculated; (g) for each pixel within the segment, its weight is multiplied by the segment mean; (h) the gaps are filled with this result; (i) the gaps are filled with this result;



Landsat-7 SLC-off: gap-filling



Landsat-7 SLC-off: gap-filling



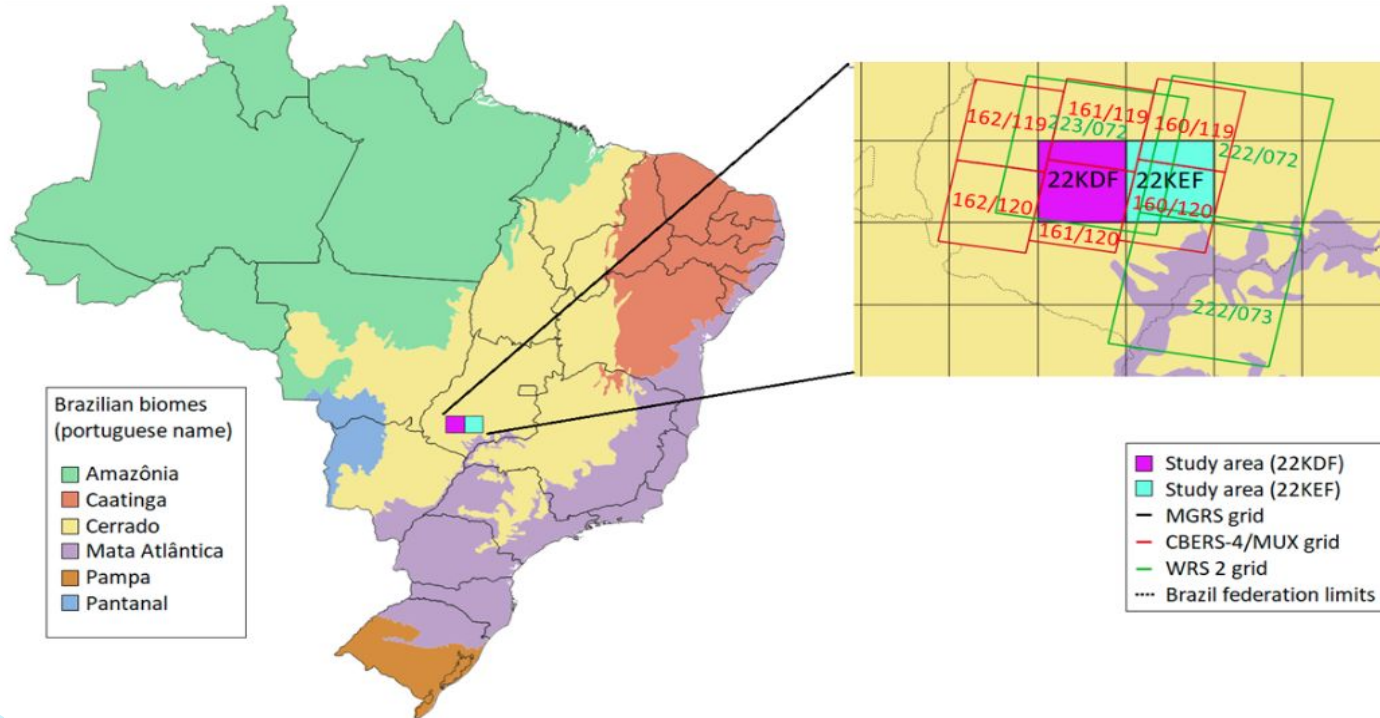
Landsat-7 SLC-off

SLC-off Landsat-7/ETM+ NIR band image (a); NIR band image from “a” filled through Maxwell et al. (2007) hierarchical multi-scale segmentation method (b) and the same image from a filled through the proposed method, which uses segmentation pixel weighting(c).



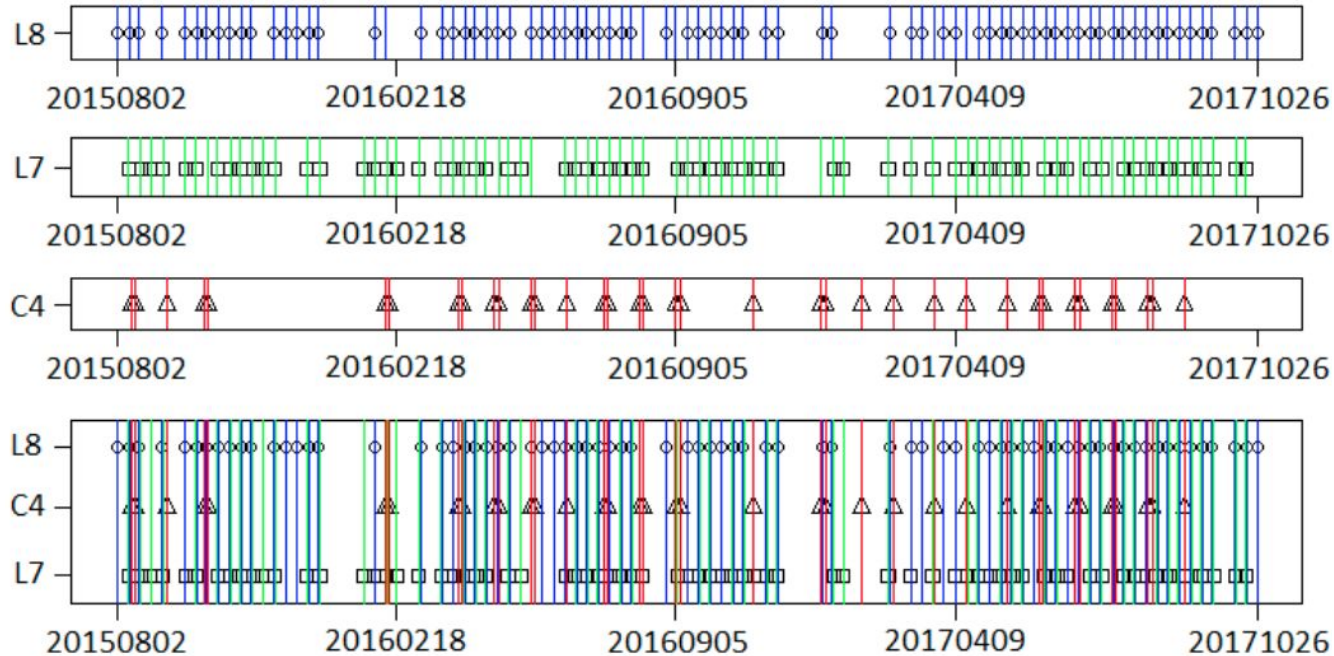
Landsat-7 SLC-off

Gap Filling of Optical Remote Sensing Multi-source data cube through multi-scale and multi-temporal segmentation (MARUJO, 2019)



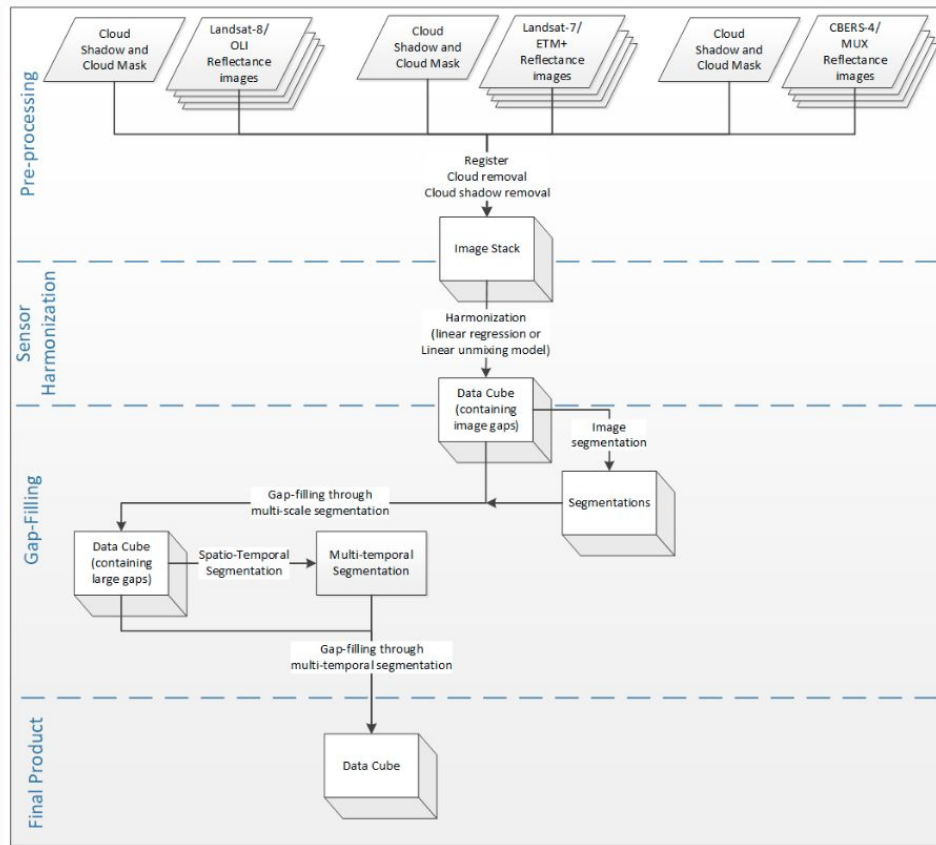
Temporal

Set of 81 Landsat-8/OLI (L8), 78 Landsat-7/ETM+ (L7) and 37 CBERS-4/MUX (C4) images in blue, green and red lines, respectively, used in the spectral harmonization in order to build a multi-source data cube (study area 22KEF), ranging from August 2015 to October 2017.



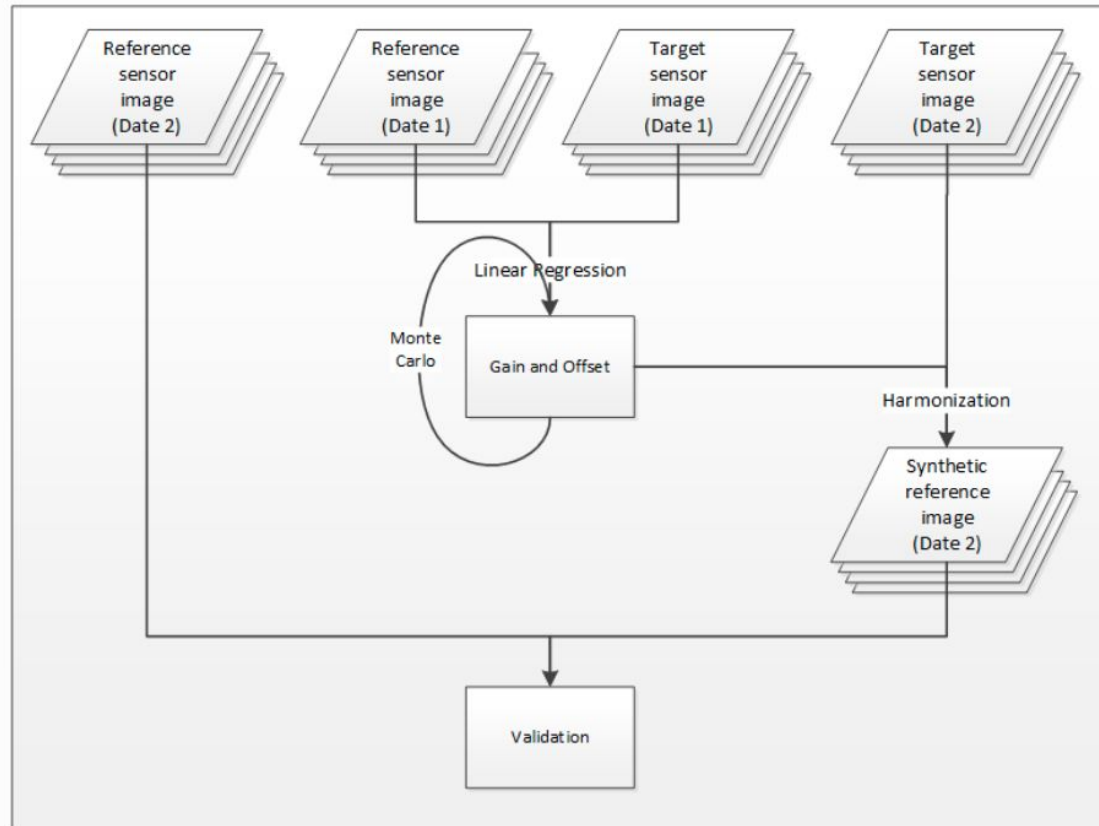
Diagram

Schematic diagram of the adopted methodology to generate multi-source RS image gap-free data cube.



Diagram

Linear Regression harmonization approach diagram.



Diagram

Spectral unmixing harmonization approach diagram.

